Université Paris-Saclay

Master 2 Optimisation

Smoothed Duality Gaps Beyond Convexity

KKT Detection, Pointwise Smoothing Thresholds,
and Proximal Escape under Affine Constraints

Internship report

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Abstract

Smoothed duality gaps provide computable merit functions in convex primal–dual optimization. We study their self-centered form for a ρ-weakly convex objective under affine constraints. If $\beta_x > \rho$ and $\beta_y > 0$, its zero set is exactly the Karush–Kuhn–Tucker (KKT) set, while the gap controls feasibility and a Lagrangian proximal fixed-point residual. For arbitrary positive smoothing, the level-proximal subdifferential characterizes the zero set. Its pointwise threshold is the least curvature globally supporting the Lagrangian at the given pair; at a KKT pair it is bounded by ρ, but may be smaller and branch-dependent. Quartic examples show that below the safe threshold, evaluating the value-defined gap requires a global inner proximal value: a center-initialized local solver may return a false zero.

For comparison, a zero global augmented-Lagrangian gap identifies a primal global minimizer and an augmented-dual maximizer, but evaluating it retains a global nonconvex inner minimization. We therefore propose a one-sided test using a different object, the hard-constrained safe proximal map $P^{\mathcal{C}}_{\lambda}$. A feasible perturbation is followed by proximal polishing; strict objective decrease rejects global optimality. Under the stated assumptions, the escape radius distinguishes global minimizers, nonglobal local minimizers, and points that are not local minimizers.

Experiments with complete analytic stationary catalogs compare the self-centered SDG, a Lagrangian proximal fixed-point residual, and a KKT residual as root-discovery objectives. In the recorded 256-start scrambled-Sobol' run on a separable weakly convex box problem, the SDG recovered all 27 analytic KKT branches; exact proximal-escape calculations then reject the exact representatives of all nonglobal branches. In the recorded shrinking-radius study, a degenerate radial example retains positive escape witnesses despite vanishing gradient and Hessian, while a flat global-minimum control analytically excludes any valid rejection witness.

Keywords: weak convexity; smoothed duality gap; level-proximal subdifferential; KKT conditions; affine constraints; optimality rejection.

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1 Introduction

A convergence theorem describes the limiting behavior of an infinite sequence, whereas a numerical method must stop at a finite iterate and decide what that iterate certifies. The choice of stopping quantity is therefore part of the mathematical meaning of an algorithm. The objective error $f(x) - f^\star$, where $f^\star$ is the optimal value, answers the desired question directly but is not generally computable. A Karush–Kuhn–Tucker (KKT) residual is usually computable, but its interpretation changes with the geometry of the problem: in convex optimization it is an optimality residual, while in nonconvex optimization it is only a stationarity residual.

For a convex–concave saddle problem, the primal–dual gap avoids this ambiguity. It is nonnegative and vanishes exactly at a saddle point, thereby combining primal optimality, dual optimality, and feasibility in one scalar. The classical gap is nevertheless poorly suited to affine equality constraints on unbounded spaces. The supremum over the multiplier is infinite whenever the current point is infeasible, and the primal comparison may also be infinite. Penalizing both comparison variables quadratically leads to the *smoothed duality gap* (SDG). Its regularized comparison problems are proximal problems, so the gap can remain finite away from feasibility and can be evaluated without knowing an optimal solution. Smooth primal–dual gap constructions were developed for convex composite minimization in [1]; the self-centered SDG was introduced in [2] and then studied systematically as a stopping criterion in [3].

The convex interpretation cannot be transferred merely by removing the word "convex". If f is proper, lower semicontinuous, and convex, every KKT pair is a saddle point of the ordinary Lagrangian and yields attained primal and dual optima with zero duality gap. Conversely, when a primal solution exists and the affine feasible set intersects $\operatorname{ri}(\operatorname{dom} f)$, strong duality and dual attainment hold and the solution admits a multiplier. Under these explicit hypotheses, the convex theory identifies

$$\text{saddle point} \quad \Longleftrightarrow \quad \text{KKT pair} \quad \Longleftrightarrow \quad \text{zero-gap primal–dual optimum}$$

see Boyd and Vandenberghe [4, Sections 5.4.2 and 5.5.3] for the smooth formulation and Rockafellar [5, Theorem 27.4] for the convex subdifferential condition under relative-interior qualification. These equivalences break down in a nonconvex problem. A KKT pair may represent a local minimum, a saddle, or a local maximum. Consequently, before using the SDG beyond convexity, one must identify its exact zero set and distinguish the statement "the gap is zero" from the much stronger statement "the point is globally optimal". A zero-based stopping rule with the wrong interpretation would accept every KKT branch, including saddles and local maxima.

1.1 Problem setting

Let $A \in \mathbb{R}^{m\times n}$ and $b \in \mathbb{R}^m$, and define the affine feasible set

$$\mathcal{C} := \{x \in \mathbb{R}^n : Ax = b\}.$$

No full-row-rank assumption is imposed on A, and we assume that $\mathcal{C} \cap \operatorname{dom} f \neq \varnothing$. We study

$$\min_{x\in\mathbb{R}^n} f(x) \quad \text{subject to} \quad Ax = b, \tag{1.1}$$

where f is proper, lower semicontinuous, and ρ-weakly convex for a valid global modulus $\rho \geq 0$. The chosen ρ need not be the smallest possible modulus. By definition, $f + \frac{\rho}{2}\|\cdot\|^2$ is convex: the negative curvature of f is globally bounded by a quadratic model. This

class includes convex functions, smooth objectives with a uniform lower Hessian bound, and many nonsmooth nonconvex models used in learning and signal recovery [6]. Controlled-concavity penalties such as SCAD [7] and MCP [8] are standard examples. Weak convexity is therefore a useful intermediate regime: it is broad enough to contain genuine nonconvexity, yet a proximal term with curvature larger than ρ makes the regularized subproblem strongly convex.

Affine constraints retain the essential primal–dual difficulty while keeping the geometry transparent. They introduce feasibility, multipliers, and tangent and normal directions, but they also permit an exact nullspace description of feasible perturbations and a dual-root representation of the constrained proximal map. Problem (1.1) is consequently a controlled setting in which both the meaning of the SDG and the cost of evaluating the proximal objects behind it can be studied. We write its ordinary Lagrangian as $\mathcal{L}(x, y) := f(x) + \langle y, Ax - b\rangle$, with multiplier $y \in \mathbb{R}^m$.

The proximal curvature creates two distinct regimes. If the primal smoothing curvature β_x is larger than ρ, the inner SDG problem is strongly convex and has a unique global minimizer; we call this the uniform safe regime. Below this threshold, the inner problem may be nonconvex and a local routine is not an oracle for the value-defined gap. At the same time, the global modulus ρ can be conservative at a fixed KKT pair. The least curvature that supports the Lagrangian globally at that pair is a pointwise quantity, and different KKT pairs of the same problem may have different thresholds. This distinction between a uniform safe regime and pointwise global support is closely related to the broader algorithmic difficulty of choosing a proximal parameter when the weak-convexity modulus is unknown [9].

The available certificates differ both in conclusion and in evaluation cost. A KKT residual is local and computable. In the safe regime, the SDG uses the Lagrangian proximal point

$$p_{\beta_x}(x, y) := \arg\min_u \left\{ \mathcal{L}(u, y) + \frac{\beta_x}{2} \|u - x\|^2 \right\}$$

together with an explicit feasibility term, and its zero set is the KKT set. The rejection stage uses a different map, the hard-constrained safe prox with $\lambda > 0$ and $\lambda\rho < 1$,

$$P_\lambda^{\mathcal{C}}(z) := \arg\min_{u \in \mathcal{C}} \left\{ f(u) + \frac{1}{2\lambda} \|u - z\|^2 \right\}.$$

A zero global augmented-Lagrangian gap instead identifies a primal global minimizer and an augmented-dual maximizer [10, 11], but requires a global infimum of a generally nonconvex augmented Lagrangian and an attaining multiplier. This leads to the intermediate question whether the hard-constrained safe prox can provide one-sided information beyond KKT without solving that global problem.

The analysis is organized around three precise questions. First, which variational condition is equivalent to zero SDG at arbitrary positive smoothing, and how does it simplify in the uniform regime $\beta_x > \rho$? This question also asks for the least curvature required at one fixed KKT pair. Second, once a stationary candidate has been found, can the hard-constrained safe proximal map exhibit a feasible point with strictly lower objective value, including when a reduced-Hessian test is inconclusive, and at what perturbation scale? Third, on problems with complete analytic stationary catalogs, how do the SDG, Lagrangian proximal fixed-point residual, and KKT residual recover KKT branches, and how does proximal escape classify the recovered candidates? These questions separate three notions that coincide in convex theory:

first-order stationarity, local minimality, global optimality.

1.2 Main results

The first main result gives an exact description of what the weakly convex SDG certifies. In the uniform safe regime, its zero set coincides with the KKT set, and the gap controls feasibility and a Lagrangian proximal fixed-point residual separately. For an arbitrary positive curvature, zero gap is characterized through the level-proximal subdifferential. This yields the pointwise threshold

$$\beta_{\mathrm{crit}}(x,y) = \sup_{\substack{u\in\operatorname{dom} f\\ u\neq x}} \frac{2[\mathcal{L}(x,y)-\mathcal{L}(u,y)]_+}{\|u-x\|^2},$$

which is bounded above by ρ at a KKT pair but may be strictly smaller. Exact quartic examples show both pair-dependent thresholds and the false zeros produced by a local inner solve.

The second main result is a post-KKT rejection framework based on the proximal-escape radius, defined as the infimum of center-perturbation radii admitting a lower proximal image. The ideal radius separates global minima, points that are not local minimizers, and nonglobal local minima. A constrained Moreau-envelope surrogate gives a smooth outer search, while tangent-space reduction and an affine dual root implement the inner prox without replacing feasibility by a finite penalty. Because the argument uses realized objective decrease rather than a quadratic model, it can also reject degenerate stationary points at which the reduced Hessian is inconclusive.

The numerical study uses complete analytic stationary sets to compare branch discovery with post-KKT rejection, including a controlled shrinking-radius test at Hessian-flat stationary points.

The work reported here comprises the analysis in Sections 3 and 4, the escape classification and affine constructions in Section 5, and the analytic and numerical studies in Section 6; the variational tools and connections to existing approaches are cited in Sections 1.3 and 2.

1.3 Related work

The SDG analysis builds on the convex theory of smoothed gaps. Tran-Dinh, Fercoq, and Cevher introduced a smooth primal–dual framework for composite convex minimization [1]. Fercoq introduced the self-centered smoothed gap [2]; Walwil and Fercoq then developed it as a stopping criterion and compared it with the KKT error, projected duality gap, and objective-gap/feasibility measure [3]. Their zero-gap interpretation uses the convex equivalence between KKT conditions, saddle points, and primal–dual optimality. Here the same ordinary Lagrangian and self-centered smoothing are examined after this equivalence fails.

Weakly convex Moreau-envelope theory provides regularity for both proximal constructions above, although their objectives and feasible domains differ. In the safe regime the envelope is continuously differentiable and its gradient is a standard stationarity measure [6]. Fixed points of the proximal map characterize stationarity rather than global optimality. Xie's adaptive proximal framework addresses the complementary problem of taking accepted steps when the global weak-convexity parameter is unknown [9]; the pointwise threshold here instead asks whether one fixed center globally minimizes its own proximal model.

The variational object connecting the SDG zero set with global proximal support is the level-proximal subdifferential. It represents proximal maps of possibly nonconvex functions as resolvents without restricting the function to the hypoconvex case [12]. Unlike a local proximal subdifferential, it imposes a quadratic support inequality against every comparison point, precisely matching the global inner infimum in the SDG. This supplies the pointwise language for the zero-set and threshold results.

Negative reduced-Hessian curvature is the classical local signature of a strict saddle, but it is silent at degenerate stationary points whose first and second derivatives vanish. Higher-order escape methods use Taylor information beyond the Hessian [13], while linear regularization can restore strict-saddle geometry in selected models [14]. For weakly convex objectives, Davis and Drusvyatskiy [15] establish avoidance of active strict saddles by proximal methods under suitable structural and initialization assumptions. The present escape test asks a different question: whether proximal polishing at perturbed centers supplies a lower feasible value for a fixed candidate, including when negative Hessian curvature is absent.

For a stronger global comparison, Rockafellar relates augmented Lagrangians to perturbation and stability [10], while Dinç Yalçın and Kasimbeyli develop zero-gap conditions through weak conjugacy and sharp augmentation [11]. Their global conclusions require the relevant stability, duality, and attainment hypotheses, and their dual functions retain a global inner optimization problem. Weak conjugacy is an abstract-convexity construction, distinct from the ρ-weak convexity used here to regularize proximal subproblems.

The algorithmic literature instead supplies appropriate comparisons for finding constrained stationary points. MEAL/LiMEAL [16], proximal augmented-Lagrangian methods [17], feasible descent and SQP schemes [18, 19], and distributed Douglas–Rachford splitting for weakly convex multi-block models [20] are designed to approach stationary points or local solutions. Efficient computation of affine-constrained proximal operators is itself a separate numerical problem; dual reduction and semismooth techniques for structured nonconvex prox subproblems are developed in [21]. These works provide algorithmic context for stationary-point computation, while the constrained proximal constructions below support post-KKT rejection.

The remainder is organized as follows. Section 2 introduces the KKT system, weak convexity and level-proximal support, followed by proximal maps, Moreau envelopes, and the SDG. Section 3 establishes the safe zero-set theorem. Section 4 gives the pointwise support characterization and the counterexamples beyond the uniform threshold. Section 5 develops the proximal-escape radius and its affine-constrained computation. Section 6 reports the numerical experiments, and Section 7 discusses their scope and concludes the report.

2 Preliminaries

The analysis uses three related levels of variational information. Limiting subdifferentials formulate local stationarity and the KKT system; level-proximal subdifferentials encode the global quadratic support required by an exact SDG value; and Moreau regularization turns weak convexity into well-posed proximal constructions. This section introduces the objects in that order.

2.1 Variational framework

2.1.1 Affine geometry and limiting stationarity

We work in finite-dimensional Euclidean spaces. The symbols $\langle\cdot,\cdot\rangle$ and $\|\cdot\|$ denote the Euclidean inner product and norm; for a matrix, $\|\cdot\|$ is the induced operator norm, whereas $\|\cdot\|_F$ is the Frobenius norm. We write

$$B_r(z) := \{u : \|u - z\| \le r\}, \qquad [t]_+ := \max\{t, 0\}, \qquad \operatorname{dist}(z, D) := \inf_{u\in D} \|z - u\| .$$

If D is nonempty, closed, and convex, its Euclidean projection is the single-valued map

$$\Pi_D(z) := \arg\min_{u\in D} \|u - z\| .$$

Existence, uniqueness, and nonexpansiveness of this projection are standard [22, Example 2.25 and Corollary 12.20].

For an extended-real-valued function $h : \mathbb{R}^n \to \mathbb{R} \cup \{+\infty\}$, its *effective domain* is $\operatorname{dom} h := \{x : h(x) < +\infty\}$. The function is proper when this set is nonempty, and it is lower semicontinuous at x when $h(x) \le \liminf_{u \to x} h(u)$. A set D is encoded by its indicator

$$\delta_D(x) := \begin{cases} 0, & x \in D, \\ +\infty, & x \notin D. \end{cases}$$

Thus minimizing h over D is the same as minimizing $h + \delta_D$ over $\mathbb{R}^n$. For a convex set D, $\operatorname{ri} D$ denotes its interior relative to its affine hull. These conventions follow Rockafellar and Wets [22, Sections 1.A–B and 2.H].

Recall the affine feasible set and the standing nonemptiness assumption:

$$\mathcal{C} := \{x \in \mathbb{R}^n : Ax = b\}, \qquad \mathcal{C} \cap \operatorname{dom} f \neq \varnothing.$$

Its tangent space is $\mathcal{T} := \ker A$, and its normal space is $\mathcal{T}^\perp = \operatorname{range}(A^\top)$. The ordinary Lagrangian is

$$\mathcal{L}(x, y) := f(x) + \langle y, Ax - b \rangle, \qquad y \in \mathbb{R}^m. \tag{2.1}$$

To specify the nonsmooth first-order condition precisely, let h be finite at x. Its regular (Fréchet) subdifferential is specified, for $v \in \mathbb{R}^n$, by

$$v \in \widehat{\partial} h(x) \quad \Longleftrightarrow \quad \liminf_{\substack{u \to x \\ u \neq x}} \frac{h(u) - h(x) - \langle v, u - x \rangle}{\|u - x\|} \ge 0. \tag{2.2}$$

The limiting subdifferential is characterized by

$$\begin{aligned} v \in \partial h(x) \quad \Longleftrightarrow \quad & \text{there exist sequences } (x_k), (v_k) \text{ such that} \\ & x_k \to x, \quad h(x_k) \to h(x), \quad v_k \in \widehat{\partial} h(x_k), \quad v_k \to v. \end{aligned} \tag{2.3}$$

Both subdifferentials are empty outside $\operatorname{dom} h$. These definitions follow Rockafellar and Wets [22, Definition 8.3]; throughout, ∂h without an accent means the limiting subdifferential. When h is continuously differentiable on a neighborhood of x, the regular and limiting subdifferentials reduce to the ordinary gradient:

$$\widehat{\partial} h(x) = \partial h(x) = \{\nabla h(x)\}.$$

See Rockafellar and Wets [22, Exercise 8.8(b)]. For a proper convex function, both subdifferentials coincide with the convex subdifferential: $v \in \partial h(x)$ if and only if $h(u) \ge h(x) + \langle v, u - x \rangle$ for every u [22, Proposition 8.12]. If s is continuously differentiable on a neighborhood of x, then the limiting subdifferential obeys the exact smooth-perturbation rule [22, Exercise 8.8(c)]:

$$\partial(h + s)(x) = \partial h(x) + \nabla s(x).$$

For a closed set D, the limiting normal cone is

$$N_D(x) := \partial \delta_D(x).$$

For convex D, this agrees with the familiar support description

$$N_D(x) = \{v : \langle v, u - x \rangle \le 0 \text{ for every } u \in D\}, \qquad x \in D,$$

and is empty when $x \notin D$; see Rockafellar and Wets [22, Theorem 6.9 and Exercise 8.14]. In particular, $N_{\mathcal{C}}(x) = \operatorname{range}(A^\top)$ at every feasible point. The limiting Fermat rule states that a local minimizer of a proper lower-semicontinuous function satisfies $0 \in \partial h(x)$; for a proper convex function this condition is also sufficient for global minimality [22, Theorem 10.1].

Definition 2.1 (KKT pair)**.** A pair $(x, y) \in \mathbb{R}^n \times \mathbb{R}^m$ is a KKT pair for equation (1.1) if

$$Ax = b, \qquad 0 \in \partial f(x) + A^\top y. \tag{2.4}$$

In a convex problem, equation (2.4) is sufficient for global primal–dual optimality by the global subgradient inequality. In a nonconvex problem it is only a first-order condition: it can hold at a local minimum, saddle, or maximum.

2.1.2 Weak convexity and level-proximal support

The central curvature hypothesis is weak convexity, in the standard sense of Davis and Drusvyatskiy [6, Section 2].

Definition 2.2 (Weak convexity)**.** A proper lower-semicontinuous function $f : \mathbb{R}^n \to \mathbb{R} \cup \{+\infty\}$ is ρ-weakly convex, for $\rho \geq 0$, if

$$x \longmapsto f(x) + \frac{\rho}{2} \|x\|^2$$

is convex.

For comparison, a proper function q is μ-strongly convex, $\mu > 0$, when $q - \mu \|\cdot\|^2/2$ is convex [22, Exercise 12.59]. Hence adding a quadratic with curvature strictly larger than ρ to a ρ-weakly convex function produces a strongly convex model and, when the model is lower semicontinuous, a unique minimizer.

For every $x \in \operatorname{dom} f$ and $v \in \partial f(x)$, weak convexity gives the global lower model

$$f(u) \geq f(x) + \langle v, u - x \rangle - \frac{\rho}{2} \|u - x\|^2, \tag{2.5}$$

for every $u \in \mathbb{R}^n$; see Davis and Drusvyatskiy [6, Lemma 2.1] and the hypoconvexity characterization in Rockafellar and Wets [22, Exercise 12.61]. At a differentiability point, one has $v = \nabla f(x)$. If $f \in C^2(\mathbb{R}^n)$, weak convexity with modulus ρ is equivalent to the uniform Hessian bound $\nabla^2 f(x) \succeq -\rho I$ for every x. Weak convexity has become a standard model for nonsmooth nonconvex optimization because a sufficiently strong quadratic regularizer restores convexity; see Davis and Drusvyatskiy [6].

Weak convexity also permits an exact affine sum rule. Since $f + \rho \|\cdot\|^2/2$ and $\delta_{\mathcal{C}}$ are convex, the qualification

$$\operatorname{ri}(\operatorname{dom} f) \cap \mathcal{C} \neq \varnothing$$

gives

$$\partial(f + \delta_{\mathcal{C}})(x) = \partial f(x) + N_{\mathcal{C}}(x) = \partial f(x) + \operatorname{range}(A^\top), \qquad x \in \mathcal{C} \cap \operatorname{dom} f. \tag{2.6}$$

This is the convex sum rule of Rockafellar [5, Theorem 23.8], combined with the smooth-perturbation rule above. Consequently, equation (2.4) is equivalent to $0 \in \partial(f + \delta_{\mathcal{C}})(x)$ under this qualification. The condition is automatic when f is finite everywhere. A KKT pair always implies constrained stationarity; the qualification is needed for the converse existence of a multiplier.

The lower model in equation (2.5) uses the same global quadratic support that governs an exact nonconvex SDG value. The appropriate pointwise object is the *level-proximal subdifferential* introduced by Wang and Wang [12, Definition 3.2]. We parameterize it by curvature rather than proximal stepsize: $\beta = 1/\lambda$ for $\beta > 0$. At $\beta = 0$, we use the global affine-support endpoint identified in Wang and Wang [12, Proposition 3.6(v)].

Definition 2.3 (Level-proximal subdifferential)**.** Let $h : \mathbb{R}^n \to \mathbb{R} \cup \{+\infty\}$ be proper, let $x \in \operatorname{dom} h$, and let $\beta \geq 0$. For $g \in \mathbb{R}^n$,

$$g \in \partial_{\mathrm{p}}^{\beta} h(x) \quad \Longleftrightarrow \quad h(u) \geq h(x) + \langle g, u - x \rangle - \frac{\beta}{2} \|u - x\|^2 \quad \text{for every } u \in \mathbb{R}^n. \tag{2.7}$$

Unlike the classical proximal subdifferential, which requires a quadratic support inequality only locally, equation (2.7) is global. Increasing β weakens the support inequality, and hence

$$0 \leq \beta_1 \leq \beta_2 \quad \Longrightarrow \quad \partial_{\mathrm{p}}^{\beta_1} h(x) \subseteq \partial_{\mathrm{p}}^{\beta_2} h(x). \tag{2.8}$$

This monotonicity is proved in Wang and Wang [12, Proposition 3.6(ii)]. The comparison with limiting stationarity below follows from Wang and Wang [12, Proposition 3.4(ii), Proposition 3.6(iv), and Theorem 3.9(i)]; the endpoint $\beta = 0$ is covered by Wang and Wang [12, Proposition 3.6(v)], and the boundary $\beta = \rho$ follows from the global weak-convexity model [6, Lemma 2.1].

Lemma 2.4 (Level-proximal and limiting subdifferentials)**.** *Let h be proper and lower semicontinuous.*

(i) For every $\beta \geq 0$, $\partial_{\mathrm{p}}^{\beta} h(x) \subseteq \partial h(x)$.

(ii) If h is ρ-weakly convex, then, for every $\beta \geq \rho$,

$$\partial_{\mathrm{p}}^{\beta} h(x) = \partial h(x). \tag{2.9}$$

Thus the level-proximal condition coincides with limiting stationarity throughout the uniform safe-curvature regime. Below that regime it can be strictly stronger, because it records the least global quadratic support needed at the particular point. The proximal representation of this support condition is introduced next.

2.2 Proximal and primal–dual constructions

2.2.1 Moreau envelopes, proximal maps, and conjugacy

For $\lambda > 0$, the Moreau envelope and proximal mapping of Rockafellar and Wets [22, Definition 1.22] are

$$P_\lambda(z) = \operatorname{prox}_{\lambda f}(z) := \operatorname*{arg\,min}_{u \in \mathbb{R}^n} \left\{ f(u) + \frac{1}{2\lambda} \|u - z\|^2 \right\}, \tag{2.10}$$

$$\mathcal{E}_\lambda(z) := \inf_{u \in \mathbb{R}^n} \left\{ f(u) + \frac{1}{2\lambda} \|u - z\|^2 \right\}. \tag{2.11}$$

In general, $P_\lambda(z)$ may be set valued. If f is ρ-weakly convex and $\lambda\rho < 1$, the proximal objective is $(\lambda^{-1} - \rho)$-strongly convex. The proximal mapping is then single valued, and the Moreau envelope is continuously differentiable with

$$\nabla \mathcal{E}_\lambda(z) = \frac{1}{\lambda}(z - \operatorname{prox}_{\lambda f}(z)). \tag{2.12}$$

Thus a proximal fixed point is exactly a zero of $\nabla \mathcal{E}_\lambda$, and $\|\nabla \mathcal{E}_\lambda(z)\|$ is used as a smooth stationarity measure; see Davis and Drusvyatskiy [6, Lemma 2.2].

The level-proximal support condition has an exact proximal representation. For a proper lower-semicontinuous function h, let $\operatorname{prox}_{\lambda h}(z)$ denote the proximal argmin in equation (2.10) with f replaced by h. For every $\lambda > 0$,

$$p \in \operatorname{prox}_{\lambda h}(z) \quad \Longleftrightarrow \quad \frac{z - p}{\lambda} \in \partial_{\mathrm{p}}^{1/\lambda} h(p), \tag{2.13}$$

which is the level-proximal resolvent relation of Wang and Wang [12, Proposition 3.4(ii)]. Unlike a first-order optimality condition, equation (2.13) characterizes a global proximal minimizer even when the proximal objective is nonconvex.

For the affine problem, it is useful to absorb feasibility into the objective:

$$P_\lambda^{\mathcal{C}}(z) := \arg\min_{u\in\mathcal{C}} \left\{ f(u) + \frac{1}{2\lambda}\|u-z\|^2 \right\}, \tag{2.14}$$

$$\mathcal{E}_\lambda^{\mathcal{C}}(z) := \inf_{u\in\mathcal{C}} \left\{ f(u) + \frac{1}{2\lambda}\|u-z\|^2 \right\}. \tag{2.15}$$

Since $\delta_{\mathcal{C}}$ is convex, $f+\delta_{\mathcal{C}}$ remains ρ-weakly convex. Therefore the same safe condition $\lambda\rho<1$ yields uniqueness on $\mathcal{C}$. Applying standard weakly convex Moreau calculus to $h = f+\delta_{\mathcal{C}}$ gives the following result [15, Lemma 2.5]. It can equivalently be obtained by completing the square and using convex Moreau calculus [22, Theorem 2.26 and Exercise 12.61]; its fixed-point statement is the hypoconvex resolvent identity discussed by Wang and Wang [12, Theorems 3.1 and 3.7].

Lemma 2.5 (Standard constrained Moreau calculus)**.** *Suppose that $f+\delta_{\mathcal{C}}$ is proper, lower semicontinuous, and ρ-weakly convex, and let $\lambda>0$ with $\lambda\rho<1$. Then:*

(i) $P_\lambda^{\mathcal{C}}$ is single valued and

$$\left\| P_\lambda^{\mathcal{C}}(z) - P_\lambda^{\mathcal{C}}(w) \right\| \le \frac{1}{1-\lambda\rho}\|z-w\|; \tag{2.16}$$

(ii) $\mathcal{E}_\lambda^{\mathcal{C}}$ is continuously differentiable, with

$$\nabla\mathcal{E}_\lambda^{\mathcal{C}}(z) = \lambda^{-1}\left(z - P_\lambda^{\mathcal{C}}(z)\right); \tag{2.17}$$

(iii) a feasible point x satisfies $P_\lambda^{\mathcal{C}}(x)=x$ if and only if $0\in\partial(f+\delta_{\mathcal{C}})(x)$. If $\mathrm{ri}(\mathrm{dom}\, f)\cap\mathcal{C}\neq\varnothing$, this is equivalent to the existence of a multiplier y for which (x,y) is a KKT pair.

A zero constrained-envelope gradient therefore characterizes a stationary point of the constrained problem, not necessarily a global minimizer.

The affine proximal oracle developed in Section 5.3 also uses Fenchel conjugacy. The Fenchel conjugate of a proper function h is

$$h^*(q) := \sup_{u\in\mathbb{R}^n}\{\langle q,u\rangle - h(u)\}. \tag{2.18}$$

This is the Legendre–Fenchel transform of Rockafellar and Wets [22, Section 11.A]. If h is proper, lower semicontinuous, and μ-strongly convex, then h^* is continuously differentiable and ∇h^* is $1/\mu$-Lipschitz [22, Proposition 12.60]. Applied to $h(u)=\lambda f(u)+\|u\|^2/2$, this identity yields the unconstrained proximal map and the smooth dual-root potential used later.

2.2.2 The self-centered smoothed duality gap

Quadratic regularization also turns the classical primal–dual gap into a finite proximal construction. For a convex–concave Lagrangian, the unregularized gap at (x,y) is

$$\sup_{u,v}\{\mathcal{L}(x,v)-\mathcal{L}(u,y)\}.$$

It is nonnegative because $(u, v) = (x, y)$ is admissible, but it can be infinite. Quadratically penalizing the comparison variables gives the self-centered SDG of Fercoq [2, Definitions 4–5] and Walwil and Fercoq [3, Definitions 9–10]:

$$\mathcal{G}_{\beta_x,\beta_y}(x, y) := \sup_{\substack{u\in\mathbb{R}^n \\ v\in\mathbb{R}^m}} \left\{ \mathcal{L}(x, v) - \mathcal{L}(u, y) - \frac{\beta_x}{2} \|u - x\|^2 - \frac{\beta_y}{2} \|v - y\|^2 \right\}, \tag{2.19}$$

for $(x, y) \in \operatorname{dom} f \times \mathbb{R}^m$, with $\beta_x, \beta_y > 0$. This parameterization uses curvature $\beta_x = 1/\lambda$. Consequently the safe weak-convexity condition is $\beta_x > \rho$, equivalently $\lambda\rho < 1$.

In convex optimization, the SDG is a computable merit function with useful error bounds and algorithmic interpretations [3]. Gap functions have also been developed for nonconvex constrained problems through weak conjugacy and augmented Lagrangians [11]. In the present weakly convex setting, however, the inner comparison in equation (2.19) is global. Local stationarity alone therefore cannot characterize its zero set. The decomposition in the next section exposes the corresponding global support condition and brings the level-proximal calculus of equations (2.7) and (2.13) into the SDG analysis.

3 Zero sets of the weakly convex smoothed duality gap

Eliminating the dual comparison variable reveals two nonnegative defects. Their decomposition identifies the exact SDG zero set through the level-proximal subdifferential; above the uniform weak-convexity threshold it reduces to KKT equivalence and quantitative residual bounds.

3.1 Proximal decomposition of the SDG

The dual comparison variable in equation (2.19) can be eliminated exactly. Let $r(x) := Ax - b$. Completing the square gives the unique maximizer

$$v_{\beta_y}(x, y) = y + \beta_y^{-1} r(x) \tag{3.1}$$

and the value

$$\sup_v \left\{ \langle v, r(x)\rangle - \frac{\beta_y}{2} \|v - y\|^2 \right\} = \langle y, r(x)\rangle + \frac{1}{2\beta_y} \|r(x)\|^2 .$$

Consequently

$$\mathcal{G}_{\beta_x,\beta_y}(x, y) = \mathcal{L}(x, y) - \inf_u \left\{ \mathcal{L}(u, y) + \frac{\beta_x}{2} \|u - x\|^2 \right\} + \frac{1}{2\beta_y} \|Ax - b\|^2 . \tag{3.2}$$

This formula separates a self-centered proximal defect, which measures whether x is a global minimizer of a quadratically regularized Lagrangian, from an explicit squared feasibility defect. It is valid as an extended-valued identity: if the primal infimum is $-\infty$, then the proximal defect and the SDG are $+\infty$. The extended-valued formulation therefore precedes any attainment or uniqueness requirement; those properties enter in the safe regime below. Accordingly, throughout Sections 3 and 4, the SDG is understood on $\operatorname{dom} f \times \mathbb{R}^m$ as a function with values in $[0, +\infty]$.

The decomposition also fixes an important sign convention. A larger β_x means stronger proximal curvature, whereas a larger step $\lambda = 1/\beta_x$ means weaker curvature.

For later reference, introduce the value function

$$m_{\beta_x}(x, y) := \inf_u \left\{ \mathcal{L}(u, y) + \frac{\beta_x}{2} \|u - x\|^2 \right\}. \tag{3.3}$$

Then the SDG is the sum of the two nonnegative quantities

$$\mathcal{G}_{\beta_x,\beta_y}(x,y) = \underbrace{\mathcal{L}(x,y) - m_{\beta_x}(x,y)}_{\text{global self-centered proximal defect}} + \underbrace{\frac{1}{2\beta_y}\|r(x)\|^2}_{\text{feasibility defect}}. \tag{3.4}$$

The first defect uses the global value m_{β_x}. An inner routine that returns only a local model value evaluates a different numerical quantity whose zero set may differ from that characterized below. Figure 3.1 illustrates the two comparison problems and their contributions to equation (3.4).

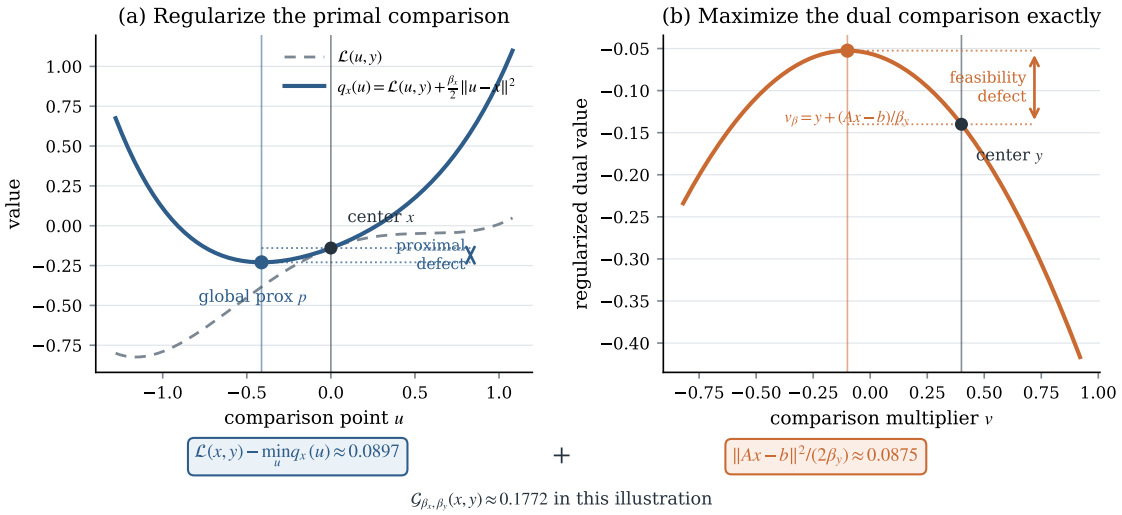


Fig. 3.1 Decomposition of a self-centered SDG evaluation. For $f(u) = u^4/4 - u^2/2$, $A = 1$, $b = 0.35$, $x = 0$, $y = 0.4$, and $(\rho, \beta_x, \beta_y) = (1, 1.8, 0.7)$, the global proximal decrease in (a) and the explicit feasibility defect in (b) sum to $\mathcal{G}_{\beta_x,\beta_y}(x,y) \approx 0.1772$.

Theorem 3.1 (Exact SDG zero set at arbitrary curvature)**.** *Let f be proper and lower semicontinuous. Fix $(x,y) \in \operatorname{dom} f \times \mathbb{R}^m$ and $\beta_x, \beta_y > 0$. Interpret the SDG as an extended-valued quantity through the global supremum in equation (2.19). Then $\mathcal{G}_{\beta_x,\beta_y}(x,y) \geq 0$, and*

$$\mathcal{G}_{\beta_x,\beta_y}(x,y) = 0 \iff Ax = b \quad \text{and} \quad -A^\top y \in \partial_{\mathrm{p}}^{\beta_x} f(x). \tag{3.5}$$

In particular, every zero of the SDG is a KKT pair.

Proof. In equation (3.3), the competitor $u = x$ gives $m_{\beta_x}(x,y) \leq \mathcal{L}(x,y)$. Thus both terms in equation (3.4) are nonnegative. If the primal infimum is $-\infty$, the first term is $+\infty$ and the gap cannot vanish. Otherwise their sum vanishes if and only if $Ax = b$ and x attains the global infimum in equation (3.3), equivalently,

$$\mathcal{L}(u,y) \geq \mathcal{L}(x,y) - \frac{\beta_x}{2}\|u - x\|^2 \qquad \text{for every } u \in \mathbb{R}^n.$$

Since

$$\mathcal{L}(u,y) - \mathcal{L}(x,y) = f(u) - f(x) + \left\langle A^\top y, u - x \right\rangle,$$

the preceding global inequality is equivalent to

$$f(u) \geq f(x) + \left\langle -A^\top y, u - x \right\rangle - \frac{\beta_x}{2}\|u - x\|^2 \qquad \text{for every } u.$$

By the definition in equation (2.7), this is exactly $-A^\top y \in \partial_{\mathrm{p}}^{\beta_x} f(x)$, proving equation (3.5). Finally, Lemma 2.4(i) gives $-A^\top y \in \partial f(x)$, so feasibility yields the KKT conditions. □

At arbitrary curvature, Theorem 3.1 characterizes a zero by the exact global quadratic-support condition. The limiting-subdifferential Fermat condition captures local stationarity, whereas the converse from KKT to zero additionally requires this global support at curvature β_x.

There is an equivalent fixed-point formulation. For every $\beta_x > 0$, completion of the square gives

$$\begin{aligned}&\mathcal{L}(u,y) + \frac{\beta_x}{2}\|u-x\|^2 \\ &\quad = f(u) + \frac{\beta_x}{2}\left\|u - \left(x - \beta_x^{-1}A^\top y\right)\right\|^2 + \langle y, Ax-b\rangle - \frac{1}{2\beta_x}\left\|A^\top y\right\|^2,\end{aligned} \tag{3.6}$$

Applying the level-proximal resolvent relation equation (2.13) with $h = f$, $\lambda = \beta_x^{-1}$ and $z = x - \beta_x^{-1}A^\top y$ therefore gives

$$x \in P_{\beta_x^{-1}}\left(x - \beta_x^{-1}A^\top y\right) \quad\Longleftrightarrow\quad -A^\top y \in \partial_{\mathrm{p}}^{\beta_x} f(x). \tag{3.7}$$

Here $P_{\beta_x^{-1}}$ is allowed to be set-valued. Thus an arbitrary curvature zero is a *global* proximal fixed point, not merely a stationary point of the inner model.

3.2 Uniform smoothing and residual estimates

The arbitrary-curvature criterion is exact but still involves a global quadratic-support condition. Weak convexity supplies a uniform regime in which this condition reduces to ordinary first-order stationarity and the inner comparison problem is uniquely solvable. Fix $(x,y) \in \operatorname{dom} f \times \mathbb{R}^m$, and define

$$q_{x,y}(u) := \mathcal{L}(u,y) + \frac{\beta_x}{2}\|u-x\|^2.$$

If f is ρ-weakly convex and $\beta_x > \rho$, then $q_{x,y}$ is $(\beta_x - \rho)$-strongly convex. It is also proper, lower semicontinuous, and coercive, hence has a unique minimizer

$$p_{\beta_x}(x,y) := \arg\min_u q_{x,y}(u) = \operatorname{prox}_{\beta_x^{-1}f}\left(x - \beta_x^{-1}A^\top y\right). \tag{3.8}$$

The corresponding optimality condition is

$$\beta_x(x - p_{\beta_x}(x,y)) \in \partial f(p_{\beta_x}(x,y)) + A^\top y. \tag{3.9}$$

Writing $p = p_{\beta_x}(x,y)$, the value formula equation (3.2) becomes

$$\begin{aligned}\mathcal{G}_{\beta_x,\beta_y}(x,y) &= \mathcal{L}(x,y) - \mathcal{L}(p,y) - \frac{\beta_x}{2}\|p-x\|^2 + \frac{1}{2\beta_y}\|Ax-b\|^2 \\ &= f(x) - f(p) + \langle y, A(x-p)\rangle - \frac{\beta_x}{2}\|p-x\|^2 + \frac{1}{2\beta_y}\|Ax-b\|^2.\end{aligned} \tag{3.10}$$

Thus one safe SDG evaluation requires one globally well-posed Lagrangian proximal problem; neither an optimal primal point nor the value $f^\star$ is an input. Its evaluation cost is determined by this strongly convex inner solve.

Theorem 3.2 (Safe-regime KKT equivalence and residual bound)**.** *Let f be proper, lower semicontinuous, and ρ-weakly convex. Let $(x,y) \in \operatorname{dom} f \times \mathbb{R}^m$, $\beta_x > \rho$, and $\beta_y > 0$. Then*

$$\mathcal{G}_{\beta_x,\beta_y}(x,y) \geq \frac{\beta_x - \rho}{2}\|x - p_{\beta_x}(x,y)\|^2 + \frac{1}{2\beta_y}\|Ax-b\|^2. \tag{3.11}$$

In particular,

$$\mathcal{G}_{\beta_x,\beta_y}(x,y) = 0 \quad\Longleftrightarrow\quad (x,y) \text{ is a KKT pair.} \tag{3.12}$$

Proof. Since $f + \rho\|\cdot\|^2/2$ is convex, adding the affine term $\langle y, A\cdot - b\rangle$ and the quadratic centered at x shows that $q_{x,y}$ is $(\beta_x - \rho)$-strongly convex. Its unique minimizer p satisfies $0 \in \partial q_{x,y}(p)$. The strong-convexity inequality at this minimizer gives

$$q_{x,y}(u) \geq q_{x,y}(p) + \frac{\beta_x - \rho}{2}\|u - p\|^2 \qquad \text{for every } u.$$

Taking $u = x$ and combining the result with equation (3.2) proves equation (3.11). If the gap is zero, then $x = p$ and $Ax = b$. Substitution into equation (3.9) gives $0 \in \partial f(x) + A^\top y$.

Conversely, suppose (x, y) is KKT. Weak convexity and $-A^\top y \in \partial f(x)$ give

$$\mathcal{L}(u, y) \geq \mathcal{L}(x, y) - \frac{\rho}{2}\|u - x\|^2.$$

Consequently

$$q_{x,y}(u) \geq q_{x,y}(x) + \frac{\beta_x - \rho}{2}\|u - x\|^2 \qquad \text{for every } u.$$

Hence x is the unique global minimizer p. Feasibility makes both defects in equation (3.4) zero and proves the converse. □

Remark 3.3 (The boundary). At $\beta_x = \rho > 0$, the KKT-to-zero argument still yields a global quadratic support and therefore zero gap at a KKT pair. However, an arbitrary inner problem is only convex, not strongly convex: it may be noncoercive or have multiple minimizers. We therefore reserve *safe computational regime* for the strict condition $\beta_x > \rho$, and treat the closed-boundary zero-set statement through the pointwise criterion in Section 4.

Theorem 3.2 extends the self-centered convex SDG zero-set principle of Fercoq [2], Walwil and Fercoq [3] to weak convexity, but the meaning of the zero set changes. In the convex case, KKT pairs are saddle points and therefore globally primal–dual optimal. In the weakly convex case, the same equality accepts every KKT type.

For instance, with $f(t) = -t^2$, the point $t = 0$ is a strict local maximum and a stationary point. For any $0 < \lambda < 1/2$,

$$\operatorname{prox}_{\lambda f}(0) = 0,$$

so both the Moreau-gradient residual and the corresponding safe SDG vanish at the maximum. This behavior is intrinsic to a first-order detector and motivates the post-KKT rejection test of Section 5.

Beyond this exact zero-set interpretation, the lower bound equation (3.11) yields the computable estimates

$$\|x - p_{\beta_x}(x, y)\| \leq \sqrt{\frac{2\mathcal{G}_{\beta_x,\beta_y}(x, y)}{\beta_x - \rho}}, \tag{3.13}$$

$$\|Ax - b\| \leq \sqrt{2\beta_y \mathcal{G}_{\beta_x,\beta_y}(x, y)}. \tag{3.14}$$

The first bound also controls stationarity at the proximal comparison point. Indeed, equation (3.9) yields

$$\operatorname{dist}\Big(0, \partial f(p_{\beta_x}(x, y)) + A^\top y\Big) \leq \beta_x \sqrt{\frac{2\mathcal{G}_{\beta_x,\beta_y}(x, y)}{\beta_x - \rho}}. \tag{3.15}$$

Feasibility at that same comparison point follows from the triangle inequality:

$$\|Ap_{\beta_x}(x, y) - b\| \leq \sqrt{2\beta_y \mathcal{G}_{\beta_x,\beta_y}(x, y)} + \|A\| \sqrt{\frac{2\mathcal{G}_{\beta_x,\beta_y}(x, y)}{\beta_x - \rho}}. \tag{3.16}$$

Moreover, the displacement of the maximizing dual comparison point in equation (3.1) satisfies

$$\left\|v_{\beta_y}(x,y)-y\right\| \le \sqrt{\frac{2\mathcal{G}_{\beta_x,\beta_y}(x,y)}{\beta_y}}. \tag{3.17}$$

Thus a small value simultaneously controls a primal proximal displacement, stationarity at the proximal point, primal feasibility, and the dual comparison displacement. These are stationarity and feasibility bounds; an estimate of $f(x)-f^\star$ in the nonconvex setting requires additional global assumptions.

Strong convexity also converts an inner stopping residual into a certified gap enclosure.

Lemma 3.4 (Inexact safe-prox enclosure)**.** *Under the assumptions of Theorem 3.2, set $\mu := \beta_x - \rho$, let $p = \arg\min q_{x,y}$, and suppose that $\widehat{p} \in \operatorname{dom} q_{x,y}$ satisfies*

$$\operatorname{dist}(0, \partial q_{x,y}(\widehat{p})) \le \varepsilon.$$

Then

$$\|\widehat{p}-p\| \le \frac{\varepsilon}{\mu}, \qquad 0 \le q_{x,y}(\widehat{p}) - q_{x,y}(p) \le \frac{\varepsilon^2}{2\mu}. \tag{3.18}$$

If

$$\widehat{\mathcal{G}}_{\beta_x,\beta_y}(x,y) := \mathcal{L}(x,y) - q_{x,y}(\widehat{p}) + \frac{1}{2\beta_y}\|Ax-b\|^2, \tag{3.19}$$

then the exact gap belongs to the computable interval

$$\mathcal{G}_{\beta_x,\beta_y}(x,y) \in \left[\max\left\{0, \widehat{\mathcal{G}}_{\beta_x,\beta_y}(x,y)\right\}, \widehat{\mathcal{G}}_{\beta_x,\beta_y}(x,y) + \frac{\varepsilon^2}{2(\beta_x-\rho)}\right]. \tag{3.20}$$

Proof. The function $q_{x,y}$ is proper, lower semicontinuous, and μ-strongly convex. Its subdifferential at $\widehat{p}$ is a nonempty closed convex set, so the distance to zero is attained. Choose $\widehat{v} \in \partial q_{x,y}(\widehat{p})$ with $\|\widehat{v}\| \le \varepsilon$. Strong monotonicity of its subdifferential, together with $0 \in \partial q_{x,y}(p)$, gives

$$\mu\|\widehat{p}-p\|^2 \le \langle \widehat{v}, \widehat{p}-p\rangle \le \varepsilon\|\widehat{p}-p\|,$$

proving the distance estimate. The strong-convexity inequality at $\widehat{p}$ also gives

$$q_{x,y}(\widehat{p}) - q_{x,y}(p) \le \langle \widehat{v}, \widehat{p}-p\rangle - \frac{\mu}{2}\|\widehat{p}-p\|^2 \le \frac{\varepsilon^2}{2\mu}.$$

Finally, $\mathcal{G}_{\beta_x,\beta_y} - \widehat{\mathcal{G}}_{\beta_x,\beta_y} = q_{x,y}(\widehat{p}) - q_{x,y}(p)$, and the exact gap is nonnegative. □

The factor $1/2$ is sharp: for $q(u) = \mu u^2/2$, the point $\widehat{p} = \varepsilon/\mu$ has residual ε and value error $\varepsilon^2/(2\mu)$.

The residual in Lemma 3.4 is the full subdifferential residual. For an extended-valued box model it therefore includes the normal cone of the box, not only the gradient of the smooth quadratic part. This is the residual required when the enclosure is applied to an inexact box proximal evaluation.

3.3 Stationary geometry of the SDG

In the safe regime, the SDG is nonnegative and every KKT pair has value zero. Thus, if the KKT set is nonempty,

$$\operatorname*{arg\,min}_{(x,y)\in \operatorname{dom} f\times\mathbb{R}^m} \mathcal{G}_{\beta_x,\beta_y}(x,y) = \left\{(x,y) : Ax=b,\ 0 \in \partial f(x) + A^\top y\right\}, \qquad \beta_x > \rho. \tag{3.21}$$

Thus every KKT branch is a global-minimum branch of the SDG, including branches whose primal components do not minimize the original problem locally. Local minimization of this merit need not discover all such branches.

For a finite-valued smooth objective, uniqueness of the safe proximal point also permits differentiation of the value function. This additional regularity is not needed for the nonsmooth zero-set theorem.

Proposition 3.5 (Gradient of the safe SDG)**.** *Suppose that $f : \mathbb{R}^n \to \mathbb{R}$ is continuously differentiable and ρ-weakly convex, with $\beta_x > \rho$ and $\beta_y > 0$. Set $p = p_{\beta_x}(x, y)$ as in equation (3.8). Then the SDG is continuously differentiable and*

$$\nabla_x \mathcal{G}_{\beta_x,\beta_y}(x, y) = \nabla f(x) + A^\top y - \beta_x(x - p) + \frac{1}{\beta_y} A^\top (Ax - b), \tag{3.22}$$

$$\nabla_y \mathcal{G}_{\beta_x,\beta_y}(x, y) = A(x - p). \tag{3.23}$$

Proof. Recall the strongly convex value function $m_{\beta_x}(x, y)$ from equation (3.3). The inner objectives have the common strong-convexity modulus $\beta_x - \rho > 0$. On every bounded neighborhood of (x, y), their affine and linear parameter terms are bounded, so a common quadratic lower bound makes the family locally uniformly level-bounded. Its minimizer $p_{\beta_x}(x, y)$ is therefore unique and continuous in (x, y). Danskin's theorem for parametric minimization [22, Theorem 10.58], applied to the two parameters of the inner objective, gives

$$\nabla_x m_{\beta_x}(x, y) = \beta_x(x - p), \qquad \nabla_y m_{\beta_x}(x, y) = Ap - b.$$

Differentiating

$$\mathcal{G}_{\beta_x,\beta_y}(x, y) = \mathcal{L}(x, y) - m_{\beta_x}(x, y) + \frac{1}{2\beta_y} \|Ax - b\|^2$$

now yields equations (3.22) and (3.23). Continuity follows from continuity of ∇f and of the safe proximal map. □

The box-constrained benchmark uses an extended-valued objective, so its smooth search objective is obtained by extending the SDG off the box rather than by differentiating the indicator. The following proposition records the precise extension used in Section 6.

Proposition 3.6 (Smooth finite extension for a convex side set)**.** *Let $B \subset \mathbb{R}^n$ be nonempty, closed, and convex, and let $F : \mathbb{R}^n \to \mathbb{R}$ be continuously differentiable and ρ-weakly convex. Suppose $\beta_x > \rho$ and $\beta_y > 0$, and set*

$$p_B(x, y) := \operatorname*{arg\,min}_{u \in B} \left\{ F(u) + \langle y, Au - b \rangle + \frac{\beta_x}{2} \|u - x\|^2 \right\}. \tag{3.24}$$

The following finite-valued function is continuously differentiable on $\mathbb{R}^n \times \mathbb{R}^m$:

$$\begin{aligned} \widetilde{\mathcal{G}}_{B,\beta_x,\beta_y}(x, y) := F(x) + \langle y, Ax - b \rangle \\ - \min_{u \in B} \left\{ F(u) + \langle y, Au - b \rangle + \frac{\beta_x}{2} \|u - x\|^2 \right\} + \frac{1}{2\beta_y} \|Ax - b\|^2 . \end{aligned} \tag{3.25}$$

Its gradients are

$$\nabla_x \widetilde{\mathcal{G}}_{B,\beta_x,\beta_y}(x, y) = \nabla F(x) + A^\top y - \beta_x(x - p_B(x, y)) + \beta_y^{-1} A^\top (Ax - b), \tag{3.26}$$

$$\nabla_y \widetilde{\mathcal{G}}_{B,\beta_x,\beta_y}(x, y) = A(x - p_B(x, y)). \tag{3.27}$$

On $B \times \mathbb{R}^m$, it agrees with the SDG associated with $f = F + \delta_B$.

Proof. Set $h = F + \delta_B$, $\lambda = \beta_x^{-1}$, and $z = x - \lambda A^\top y$. The function h is proper, lower semicontinuous, and ρ-weakly convex. Completing the square as in equation (3.6) gives

$$\widetilde{\mathcal{G}}_{B,\beta_x,\beta_y}(x,y) = F(x) - \mathcal{E}_\lambda^h(z) + \frac{\lambda}{2}\left\|A^\top y\right\|^2 + \frac{1}{2\beta_y}\|Ax-b\|^2,$$

where $\mathcal{E}_\lambda^h$ denotes the Moreau envelope of h. The safe Moreau calculus in Section 2 yields $p_B(x,y) = \operatorname{prox}_{\lambda h}(z)$ and $\nabla \mathcal{E}_\lambda^h(z) = \lambda^{-1}(z - p_B(x,y))$. The displayed expression is therefore continuously differentiable; applying the chain rule proves equations (3.26) and (3.27). Finally, if $x \in B$, substituting $f = F + \delta_B$ in equation (3.2) restricts the inner infimum to B and gives exactly equation (3.25). □

Outside B, $\widetilde{\mathcal{G}}_{B,\beta_x,\beta_y}$ is used only as a differentiable extension. Its interpretation as a nonnegative SDG, and the corresponding zero-set statements, are invoked only on $B \times \mathbb{R}^m$.

In the finite-valued smooth setting of Proposition 3.5, a KKT pair satisfies $p = x$, $Ax = b$, and $\nabla f(x) + A^\top y = 0$, so both gradients vanish. The converse from *zero value* is Theorem 3.2; the converse from *zero gradient* is false in general, since merit stationarity requires only

$$A(x-p) = 0, \qquad \nabla f(x) + A^\top y - \beta_x(x-p) + \beta_y^{-1}A^\top(Ax-b) = 0. \tag{3.28}$$

The first equation does not force the tangent proximal displacement to vanish. An explicit example is the unconstrained double well $f(t) = t^4/4 - t^2/2$, with $\rho = 1$. Take

$$\beta = \frac{30}{19} > 1, \qquad x = \frac{2}{\sqrt{19}}, \qquad p = \frac{3}{\sqrt{19}}.$$

Since $f'(p) = f'(x) = -30/(19\sqrt{19}) = \beta(x-p)$, strong convexity makes p the unique proximal comparison point, while

$$\mathcal{G}'_\beta(x) = f'(x) - \beta(x-p) = 0, \qquad \mathcal{G}_\beta(x) = f(x) - f(p) - \frac{\beta}{2}(p-x)^2 = \frac{65}{1444} > 0.$$

Thus even a smooth safe SDG can have a positive-valued stationary point. Experimental endpoints are consequently accepted only after independent recomputation of feasibility, KKT, and Lagrangian proximal fixed-point residuals; the merit value is independently recomputed and recorded separately.

Once the strongly convex Lagrangian proximal point p has been computed, (3.22)–(3.23) require no derivative of the proximal map. A raw KKT merit avoids the proximal solve, whereas a squared Lagrangian proximal fixed-point residual uses the same inner point but has a different landscape. The complete-catalog experiments in Section 6.1 compare their branch recovery.

3.4 Comparison with augmented-duality certificates

For comparison with the zero-set result, following Rockafellar [10], define for $\sigma \geq 0$

$$\mathcal{L}_\sigma(u,y) := f(u) + \langle y, Au-b\rangle + \frac{\sigma}{2}\|Au-b\|^2. \tag{3.29}$$

The local residual

$$\mathcal{R}_\sigma(x,y) := \operatorname{dist}^2(0, \partial_x \mathcal{L}_\sigma(x,y)) + \|Ax-b\|^2 \tag{3.30}$$

vanishes exactly at KKT pairs: feasibility removes the derivative of the quadratic penalty, leaving the KKT stationarity condition. Global information enters only through the value function below.

A genuinely global benchmark uses the augmented dual value

$$d_\sigma(y) := \inf_{u\in\mathbb{R}^n} \mathcal{L}_\sigma(u, y) \tag{3.31}$$

and the global augmented-Lagrangian gap

$$\mathcal{A}_\sigma(x, y) := \sup_{v\in\mathbb{R}^m} \mathcal{L}_\sigma(x, v) - d_\sigma(y) = \delta_{\{0\}}(Ax - b) + f(x) - d_\sigma(y). \tag{3.32}$$

Here the gap is formed from the quadratic augmented dual value of Rockafellar [10]. The sharp-augmentation framework of Dinç Yalçın and Kasimbeyli [11] provides a related global-duality comparison, but uses a different augmentation. Let $f^\star := \inf_{u\in\mathcal{C}} f(u)$, with $\varnothing \neq \mathcal{C}$ and $f^\star \in \mathbb{R}$. Since weak duality gives $d_\sigma(y) \le f^\star \le f(x)$ for every feasible x,

$$\mathcal{A}_\sigma(x, y) = 0 \quad\Longleftrightarrow\quad Ax = b, \qquad f(x) = f^\star, \qquad d_\sigma(y) = f^\star. \tag{3.33}$$

A zero identifies both a primal global minimizer and an attained augmented-dual optimum. Table 3.1 records the corresponding oracle requirements.

Table 3.1 Zero-based certificate hierarchy and required evaluation oracles. The global interpretation in the last row is conditional on computing a zero of the value-defined gap

Quantity	Principal oracle	Exact information
KKT residual $\mathcal{R}_\sigma$	subdifferential and feasibility evaluations	first-order stationarity
Safe SDG $\mathcal{G}_{\beta_x,\beta_y}$	one strongly convex Lagrangian proximal problem and an explicit feasibility term	KKT, with the bounds equations (3.13) and (3.14)
Global augmented-Lagrangian gap $\mathcal{A}_\sigma$	a global inner infimum $d_\sigma(y)$, and an attaining dual multiplier	primal global optimality and augmented-dual optimality

The augmented penalty is constant along $\ker A$, so evaluating $d_\sigma(y)$ can retain a global nonconvex inner problem. The safe SDG uses one well-posed Lagrangian proximal evaluation, but its zero set stops at KKT. The next section identifies the pointwise curvature at which that zero appears.

4 Pointwise smoothing thresholds

The safe regime uses ρ uniformly over all centers and multipliers, but a fixed KKT pair may require less curvature. This section gives the exact pointwise threshold and separates the value-defined condition from attainment or local solution of the inner problem.

4.1 Support curvature and the exact pointwise threshold

The global constant ρ controls every center and every multiplier, but a fixed slope at a fixed point may admit a sharper global quadratic support. Motivated by the level-proximal subdifferential in equation (2.7), define the extended-valued *support curvature*

$$\kappa_f(x \mid g) := \inf\left\{\beta \ge 0 : g \in \partial_{\mathrm{p}}^{\beta} f(x)\right\}, \qquad \inf\varnothing := +\infty. \tag{4.1}$$

Lemma 4.1 (Quotient representation of support curvature)**.** *For $x \in \operatorname{dom} f$ and $g \in \mathbb{R}^n$,*

$$\kappa_f(x \mid g) = \sup_{\substack{u\in\operatorname{dom} f\\ u\neq x}} \frac{2[f(x) - f(u) + \langle g, u - x\rangle]_+}{\|u - x\|^2}. \tag{4.2}$$

The supremum of an empty set is understood as zero. Moreover, for every finite $\beta \geq 0$,

$$g \in \partial_{\mathrm{p}}^{\beta} f(x) \quad \Longleftrightarrow \quad \beta \geq \kappa_f(x \mid g). \tag{4.3}$$

Proof. By equation (2.7), membership $g \in \partial_{\mathrm{p}}^{\beta} f(x)$ is equivalent, for every $u \in \operatorname{dom} f \setminus \{x\}$, to

$$\beta \geq \frac{2[f(x) - f(u) + \langle g, u - x\rangle]}{\|u - x\|^2}.$$

Since $\beta \geq 0$, the right-hand side can be replaced by its positive part. Taking the supremum over $u \in \operatorname{dom} f \setminus \{x\}$ proves equation (4.2); the same family of closed inequalities proves equation (4.3), including attainment at a finite threshold. □

We apply this support calculation to the stationary candidates identified by the safe SDG. At a KKT pair, feasibility holds and stationarity selects the slope $g = -A^\top y \in \partial f(x)$. The question is how far the primal smoothing can be reduced while this pair remains a zero of the gap. For this purpose, define the Lagrangian support threshold, for any $(x, y) \in \operatorname{dom} f \times \mathbb{R}^m$, by

$$\begin{aligned} \beta_{\mathrm{crit}}(x, y) &:= \kappa_f(x \mid -A^\top y) \\ &= \sup_{\substack{u \in \operatorname{dom} f \\ u \neq x}} \frac{2[\mathcal{L}(x, y) - \mathcal{L}(u, y)]_+}{\|u - x\|^2}. \end{aligned} \tag{4.4}$$

Thus $\beta_{\mathrm{crit}}(x, y)$ is precisely the least curvature for which the relevant Lagrangian slope belongs to the level-proximal subdifferential. It is a global property at the *fixed* multiplier y. A distinct, multiplier-free curvature localized to a feasible ball is introduced in equation (5.8) when proximal escape is developed.

The same number admits two equivalent optimization interpretations. For every $\beta > 0$,

$$\begin{aligned} \beta \geq \beta_{\mathrm{crit}}(x, y) \quad &\Longleftrightarrow \quad x \in \arg\min_u \left\{ \mathcal{L}(u, y) + \frac{\beta}{2} \|u - x\|^2 \right\} \\ &\Longleftrightarrow \quad x \in P_{\beta^{-1}}\left(x - \beta^{-1} A^\top y\right). \end{aligned} \tag{4.5}$$

The first equivalence is the global support inequality with all terms moved to one side; the second follows by completing the square as in equation (3.6). Consequently

$$\beta_{\mathrm{crit}}(x, y) = \inf \left\{ \beta \geq 0 : x \in \arg\min_u \left(\mathcal{L}(u, y) + \frac{\beta}{2} \|u - x\|^2 \right) \right\}. \tag{4.6}$$

The support inequality is closed in β, and the curvature monotonicity in equation (2.8) makes the admissible curvatures upward closed. Hence, when the infimum is finite, it is attained and the support/model curvatures form the closed ray $[\beta_{\mathrm{crit}}(x, y), +\infty)$, rather than merely an open sufficient range. For a feasible pair $Ax = b$, the positive curvatures producing zero SDG are exactly those satisfying $\beta_x \geq \beta_{\mathrm{crit}}(x, y)$. For $Ax \neq b$, no positive curvature produces zero SDG.

Theorem 4.2 (Pointwise zero-gap criterion)**.** *Let f be proper and lower semicontinuous. Fix $(x, y) \in \operatorname{dom} f \times \mathbb{R}^m$ and $\beta_x, \beta_y > 0$. Interpret the SDG through its global infimum. Then*

$$\mathcal{G}_{\beta_x, \beta_y}(x, y) = 0 \quad \Longleftrightarrow \quad Ax = b \quad \text{and} \quad \beta_x \geq \beta_{\mathrm{crit}}(x, y). \tag{4.7}$$

Every zero in equation (4.7) is a KKT pair. If, in addition, f is ρ-weakly convex and (x, y) is KKT, then

$$0 \leq \beta_{\mathrm{crit}}(x, y) \leq \rho. \tag{4.8}$$

Proof. By Theorem 3.1, zero gap is equivalent to feasibility and $-A^\top y \in \partial_{\mathrm{p}}^{\beta_x} f(x)$. Applying equation (4.3) and $\beta_{\mathrm{crit}}(x,y) = \kappa_f(x \mid -A^\top y)$ proves equation (4.7). The KKT implication is already part of Theorem 3.1. At a KKT pair, $-A^\top y \in \partial f(x)$; weak convexity and equation (2.9) then give $-A^\top y \in \partial_{\mathrm{p}}^{\rho} f(x)$. Therefore $\kappa_f(x \mid -A^\top y) \le \rho$, proving equation (4.8). □

For $\rho > 0$, the distinction at the boundary $\beta_x = \rho$ is exact. At a KKT pair, equation (4.8) and equation (4.7) show that the SDG vanishes for every $\beta_x \ge \rho$, including equality. Thus the strict inequality in Theorem 3.2 is not needed for the value-defined zero-set equivalence. It is needed for a *uniform computational statement*: when $\beta_x > \rho$, every inner model is strongly convex, coercive, and has a unique minimizer; when $\beta_x = \rho$, the inner model is only convex, and its behavior away from a zero can vary.

For example, let $\rho > 0$ and $f(u) = -\rho \|u\|^2/2$ with no effective constraint. At the boundary,

$$f(u) + \frac{\rho}{2}\|u - x\|^2 = -\rho\langle x, u\rangle + \frac{\rho}{2}\|x\|^2 . \tag{4.9}$$

If $x \neq 0$, the inner infimum is $-\infty$; at the stationary point $x = 0$, the model is constant and every u is a minimizer. In contrast, the quartic model in Example 4.4 has the unique boundary minimizer $u = 0$. The boundary can therefore be attained uniquely, attained nonuniquely, or fail to have a finite value, depending on the center and the recession behavior. What is lost at equality is the uniform strong-convexity guarantee, not the KKT-to-zero implication at a KKT pair.

Corollary 4.3 (Global saddle case)**.** *Let $x \in \mathcal{C} \cap \operatorname{dom} f$ and $y \in \mathbb{R}^m$. Then $\beta_{\mathrm{crit}}(x,y) = 0$ if and only if $x \in \arg\min_u \mathcal{L}(u,y)$. Equivalently, (x,y) is a global saddle pair of the ordinary Lagrangian.*

Thus the spectrum between 0 and ρ has a precise interpretation. A global saddle needs no primal smoothing. If $x \notin \arg\min_u \mathcal{L}(u,y)$, then $\beta_{\mathrm{crit}}(x,y) > 0$; here globality refers to minimization of the all-space Lagrangian at the fixed multiplier y, not to minimization of f over $\mathcal{C}$. The uniform constant ρ is only a worst-case upper bound, and exact evaluation of the pointwise threshold is a global supremum over u.

4.2 Sharp examples and inner-solve requirements

Existence of a global proximal minimizer is distinct from convexity of the inner model. A proper lower-semicontinuous function is *prox-bounded* if its Moreau envelope is finite at some center for some positive parameter. Its threshold is

$$\lambda_f^{\mathrm{pb}} := \sup\{\lambda > 0 : \mathcal{E}_\lambda(z) > -\infty \text{ for some } z \in \mathbb{R}^n\} . \tag{4.10}$$

For a proper lower-semicontinuous prox-bounded function, the standard proximal-attainment theorem states that, for every $0 < \lambda < \lambda_f^{\mathrm{pb}}$, the proximal set $P_\lambda(z)$ is nonempty and compact and the envelope is finite at every center [22, Definition 1.23 and Theorem 1.25]. The theorem therefore guarantees finiteness, attainment, and compactness; uniqueness and convexity require additional structure.

Completing the square in the SDG inner model shows that this result applies with $\lambda = \beta_x^{-1}$ and center $x - \beta_x^{-1} A^\top y$. Thus

$$\beta_x > \left(\lambda_f^{\mathrm{pb}}\right)^{-1} \quad \Longrightarrow \quad m_{\beta_x}(x,y) \text{ is finite and attained,} \tag{4.11}$$

where $1/(+\infty) = 0$. If f is ρ-weakly convex, its global quadratic minorant implies $\lambda_f^{\mathrm{pb}} \geq 1/\rho$, with the usual convention for $\rho = 0$. Hence $\beta_x > \rho$ lies strictly inside the general attainment range, but is stronger because it also guarantees a unique strongly convex inner problem.

At the threshold $\lambda = \lambda_f^{\mathrm{pb}}$, attainment must be checked separately. All quartic objectives below are bounded from below, so $\lambda_f^{\mathrm{pb}} = +\infty$: their inner minima are attained for every $\beta_x > 0$. Consequently, the false-zero phenomenon below arises from selection of a nonglobal local branch despite global attainment.

4.2.1 Sharp quartic threshold examples

Example 4.4 (Double well with a unique feasible point)**.** Consider the equality-constrained problem

$$\min_{t\in\mathbb{R}} \left\{ f(t) := \frac{1}{4}t^4 - \frac{1}{2}t^2 \right\} \qquad \text{subject to} \qquad t = 0.$$

The objective is 1-weakly convex. The unique feasible point $x = 0$, with $y = 0$, is therefore globally optimal for the constrained problem and is a KKT pair. Yet its primal proximal model is

$$\frac{1}{4}u^4 + \frac{\beta_x - 1}{2}u^2.$$

Direct minimization gives

$$\mathcal{G}_{\beta_x,\beta_y}(0,0) = \frac{1}{4}(1-\beta_x)_+^2. \tag{4.12}$$

Indeed, with $a = 1 - \beta_x$, the proximal model is $\varphi(u) = u^4/4 - au^2/2$. If $a > 0$, its global minimizers are $u = \pm\sqrt{a}$, with value $-a^2/4$; if $a \leq 0$, the center $u = 0$ is globally minimizing. Subtracting the global model value from the center value proves equation (4.12). Hence the SDG is positive for every $0 < \beta_x < 1$ and vanishes for $\beta_x \geq 1$. The pair is globally optimal *under the constraint*, but the SDG inner model compares it with infeasible points and sees the two wells. This disproves a parameter-free global-optimality interpretation.

Here $\beta_{\mathrm{crit}}(0,0) = 1 = \rho$, and at the boundary the proximal model is $u^4/4$ with a unique minimizer. This is the unique-attainment branch of the boundary alternatives described by equation (4.9). The example also separates two notions of globality: $x = 0$ is the unique global minimizer on the feasible set $\{0\}$, but it is not a global minimizer of the all-space Lagrangian. Accordingly its pointwise Lagrangian support threshold is positive.

Consider the tilted quartic

$$h(t) = \frac{1}{4}t^4 - \frac{1}{2}t^2 + 0.1t. \tag{4.13}$$

It is 1-weakly convex and has three stationary points. Their thresholds can be obtained without a numerical global optimizer. If x is any root of $h'(x) = x^3 - x + 0.1 = 0$ and $u = x + d$, exact expansion gives

$$h(x+d) - h(x) + \frac{\beta}{2}d^2 = \frac{d^2}{4}\left[d^2 + 4xd + 2(3x^2 - 1 + \beta)\right]. \tag{4.14}$$

The quadratic bracket has its minimum at $d = -2x$, where its value is $2(x^2 - 1 + \beta)$. Hence

$$\beta_{\mathrm{crit}}(x,0) = (1-x^2)_+. \tag{4.15}$$

This exact formula gives table 4.1. The global minimizer has threshold zero, the local maximum almost reaches the uniform bound, and the higher-valued local minimum needs only about one tenth of that bound.

Table 4.1 Stationary points and exact thresholds for the tilted quartic in (4.13). Numerical values are rounded.

Type	x	$h(x)$	β_{crit}
Global minimum	-1.0466805318	-0.3523860738	0
Local maximum	0.1010312579	0.0050255156	0.989792685
Higher-valued local minimum	0.9456492739	-0.1526394418	0.105747451

A two-dimensional example shows the multiplier's role in the Lagrangian support calculation:

$$f(x_1, x_2) = \sum_{i=1}^{2} \left(\frac{1}{4} x_i^4 - \frac{1}{2} x_i^2 \right), \qquad x_1 + x_2 = 1.$$

The KKT equations reduce to

$$x_1^3 - x_1 + y = 0, \qquad x_2^3 - x_2 + y = 0, \qquad x_1 + x_2 = 1,$$

and give the three pairs

$$(x_1, x_2, y) = (0, 1, 0), \quad (1, 0, 0), \quad \left(\frac{1}{2}, \frac{1}{2}, \frac{3}{8} \right).$$

Before comparing their support thresholds, consider the feasible geometry. Parametrizing the line by $x = (t, 1 - t)$ gives

$$f(t, 1 - t) = \frac{1}{2} t^2 (t - 1)^2 - \frac{1}{4}.$$

Thus $(0, 1)$ and $(1, 0)$ are global constrained minimizers, whereas $(1/2, 1/2)$ is a strict local maximum on the line. The support calculation below instead compares all-space Lagrangian values at each fixed multiplier.

At an endpoint pair, the coordinate centered at zero has model difference

$$\frac{1}{4} d^4 + \frac{\beta - 1}{2} d^2, \tag{4.16}$$

which is nonnegative for every d exactly when $\beta \geq 1$; the coordinate centered at one already globally minimizes its unregularized summand. Thus the endpoint pairs have $\beta_{\mathrm{crit}} = 1$.

At the symmetric pair, separability reduces the calculation to one coordinate. With $u_i = \frac{1}{2} + d_i$, stationarity cancels the linear term and

$$\begin{aligned} &\left(\frac{1}{4} u_i^4 - \frac{1}{2} u_i^2 + \frac{3}{8} u_i \right) - \left(\frac{1}{4} \left(\frac{1}{2} \right)^4 - \frac{1}{2} \left(\frac{1}{2} \right)^2 + \frac{3}{8} \left(\frac{1}{2} \right) \right) + \frac{\beta}{2} d_i^2 \\ &\qquad = \frac{d_i^2}{4} \left[(d_i + 1)^2 + 2\beta - \frac{3}{2} \right]. \end{aligned} \tag{4.17}$$

This expression is nonnegative for every d_i if and only if $\beta \geq 3/4$. Therefore $\beta_{\mathrm{crit}}(\frac{1}{2}, \frac{1}{2}, \frac{3}{8}) = 3/4$. At the critical value, $d_i = 0$ and $d_i = -1$ are both global minimizers of the scalar inner model, so the SDG is already zero below the uniform weak-convexity curvature $\rho = 1$, although the proximal map is set-valued. The multiplier $3/8$ is essential in this calculation because equation (4.4) uses the all-space Lagrangian and therefore includes infeasible comparison points.

The constrained maximum therefore has the smaller Lagrangian support threshold. As Figure 4.1 illustrates, the order of these thresholds does not rank the points by constrained optimality.

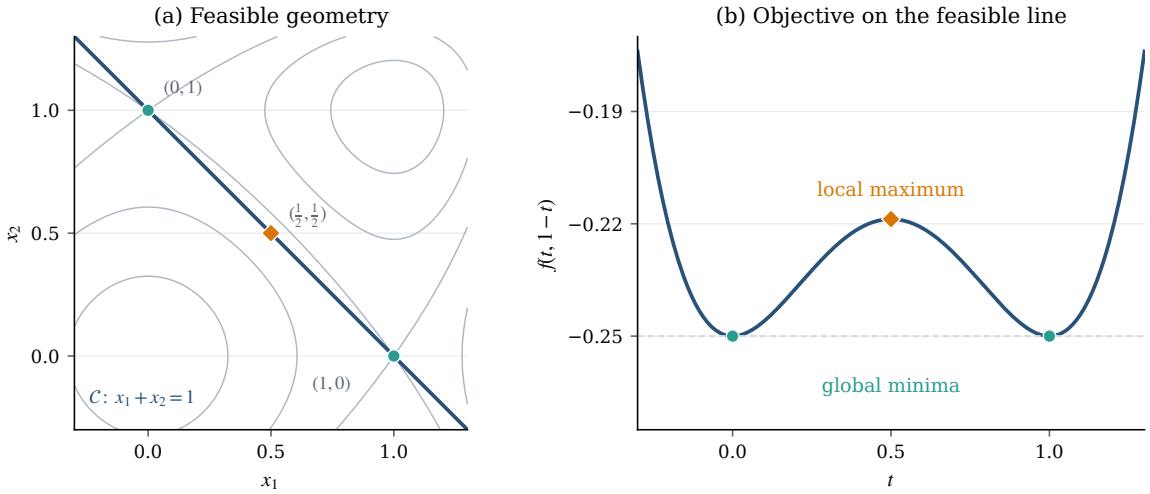


Fig. 4.1 Geometry of the affine quartic example. (a) Objective contours and the feasible line. (b) The restriction to that line has two global minima and a central local maximum; their Lagrangian support thresholds are 1, 1, and 3/4, respectively.

4.2.2 False zeros from local inner minimization

Below the safe threshold, the proximal model may be nonconvex. It is then incorrect to substitute an arbitrary local minimizer for the global infimum in equation (3.2). At the higher-valued local minimum of equation (4.13), denote the stationary point by x_{hi}, the largest root of $x^3 - x + 0.1 = 0$, with $x_{\mathrm{hi}} \approx 0.9456492739$. The center $d = 0$ remains a strict local minimizer of the proximal model for every $\beta \geq 0$, since the coefficient of d^2 in equation (4.14) is positive there. Nevertheless, when $\beta < 1 - x_{\mathrm{hi}}^2 = \beta_{\mathrm{crit}}(x_{\mathrm{hi}}, 0)$, the same factorization is negative near $d = -2x_{\mathrm{hi}}$, so the center is not globally minimizing.

The nonzero stationary displacements of the proximal model are available explicitly:

$$d_{\pm}(\beta) = \frac{-3x_{\mathrm{hi}} \pm \sqrt{4 - 3x_{\mathrm{hi}}^2 - 4\beta}}{2}. \tag{4.18}$$

The branch $d_{-}(\beta)$ gives the lower well throughout the tested range. A center-initialized local solve instead remains at $d = 0$ for $\beta_x = 0.05, 0.08, 0.10$ and therefore reports computed gap zero. Evaluation of the explicit global branch gives respectively

$$0.10272358, \qquad 0.04670322, \qquad 0.01031227.$$

The value-defined SDG remains positive at these parameters, whereas the local-inner approximation returns a false zero. The discrepancy is a branch-selection error even though the quartic model is coercive and its global minimum is attained. Figure 4.2 juxtaposes the pair-dependent exact thresholds in panel (a) with this inner-solver failure in panel (b).

These examples leave a different question after KKT detection: can a stationary candidate be rejected using a feasible point of lower objective value? Reducing β_x does not answer this question. A positive gap then detects failure of a global quadratic support for the all-space Lagrangian, which can occur even at a constrained global minimizer. Moreover, evaluating that gap below the safe regime may require a nonconvex global inner solve.

Instead, we keep proximal subproblems in the strongly convex regime and change what they compare. In Section 5, the candidate x is held fixed, the center is perturbed along the feasible set, and the distinct map $P_\lambda^{\mathcal{C}}$ imposes the affine constraint inside each proximal subproblem. Its output can therefore be compared directly with $f(x)$. The resulting escape

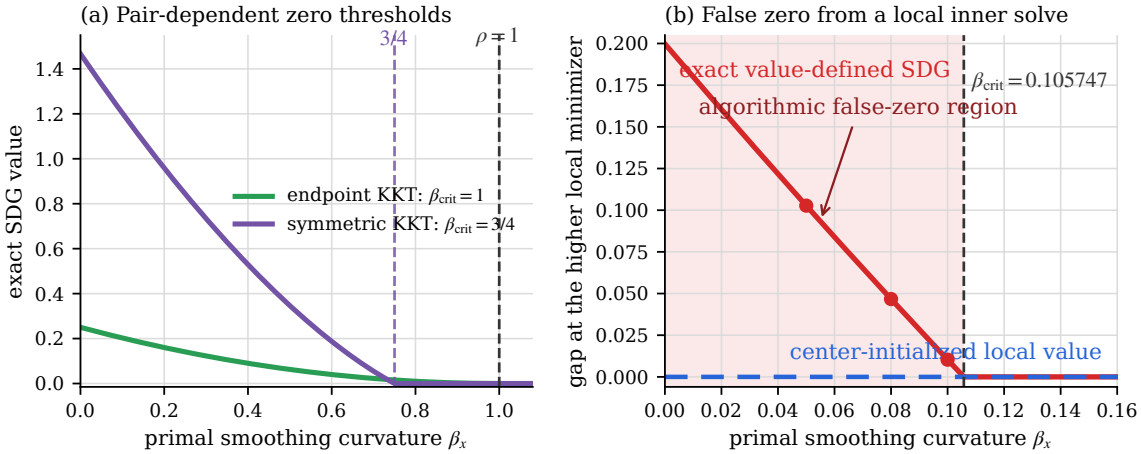


Fig. 4.2 Pointwise SDG thresholds and failure of a local inner solve. (a) The exact gaps at the endpoint and symmetric KKT pairs vanish at $\beta_x = 1$ and $3/4$, respectively. (b) At the higher-valued local minimizer of the tilted quartic, the value-defined gap is positive for $\beta_x < \beta_{\mathrm{crit}}$, whereas the center-initialized local branch returns zero.

radius measures the perturbation needed to obtain feasible descent, rather than the curvature needed to preserve a Lagrangian support.

5 Proximal-escape radii under affine constraints

After the SDG identifies a KKT candidate, proximal escape turns to a second safe proximal construction: the hard-constrained proximal map of $f + \delta_{\mathcal{C}}$. Its outputs are feasible by construction, unlike the Lagrangian proximal comparison defining the SDG. The escape radius measures the center-perturbation scale at which proximal polishing produces a strictly lower objective value.

5.1 Escape radius and its classification

Throughout this section, $x \in \mathcal{C} \cap \operatorname{dom} f$, $\lambda > 0$, $\lambda\rho < 1$, and $P_\lambda^{\mathcal{C}}$ denotes the unique constrained proximal map in equation (2.14). Center perturbations belong to $\mathcal{T} = \ker A$, so $x + d$ satisfies the affine equality but need not belong to $\operatorname{dom} f$. For example, a center may lie outside an implicit box, whereas its proximal output must belong to $\mathcal{C} \cap \operatorname{dom} f$. For $\delta \geq 0$, define the best polished value

$$\Phi_{\lambda,\delta}(x) := \inf_{\substack{d \in \mathcal{T} \\ \|d\| \leq \delta}} f\left(P_\lambda^{\mathcal{C}}(x + d)\right) \tag{5.1}$$

and its associated decrease

$$\mathcal{R}_{\lambda,\delta}(x) := f(x) - \Phi_{\lambda,\delta}(x). \tag{5.2}$$

The quantity $\mathcal{R}_{\lambda,\delta}(x)$ is nonnegative and nondecreasing in δ. It is nonnegative because the unperturbed center is admissible and proximal optimality gives $f(P_\lambda^{\mathcal{C}}(x)) \leq f(x)$; it is monotone because the set of admissible centers grows with the radius.

Definition 5.1 (Proximal-escape radius)**.** The ideal proximal-escape radius of x is

$$r_\lambda^{\mathrm{esc}}(x) := \inf\{\delta > 0 : \mathcal{R}_{\lambda,\delta}(x) > 0\}, \tag{5.3}$$

with the convention $\inf \varnothing = +\infty$.

When $r_\lambda^{\mathrm{esc}}(x) = 0$, arbitrarily small center perturbations yield a lower polished value. A finite positive radius gives a positive neighborhood $0 < \delta < r_\lambda^{\mathrm{esc}}(x)$ on which the polished decrease is zero, followed by positive decrease at every larger radius. An infinite radius means that no feasible center produces a lower proximal output.

Monotonicity makes the threshold interpretation precise. If $r = r_\lambda^{\mathrm{esc}}(x)$, then $\mathcal{R}_{\lambda,\delta}(x) = 0$ for every $0 < \delta < r$. If $r < +\infty$, then $\mathcal{R}_{\lambda,\delta}(x) > 0$ for every $\delta > r$. No conclusion about the boundary follows from monotonicity alone, but the standing assumptions provide the missing regularity. Indeed,

$$f(P_\lambda^{\mathcal{C}}(z)) = \mathcal{E}_\lambda^{\mathcal{C}}(z) - \frac{1}{2\lambda}\left\|P_\lambda^{\mathcal{C}}(z) - z\right\|^2$$

is continuous because the envelope is continuous and $P_\lambda^{\mathcal{C}}$ is Lipschitz. Hence, for every finite δ, both this function and the envelope attain their minima on the compact set $D_\delta := \mathcal{T} \cap B_\delta(0)$. To justify continuity of the outer value, write

$$h(d) := f(P_\lambda^{\mathcal{C}}(x+d)), \qquad v(\delta) := \min_{d \in D_\delta} h(d).$$

If $\delta_k \to \delta > 0$, let $d^\star$ minimize h on D_δ and set $d_k = \min\{1, \delta_k/\delta\}d^\star$. Then $d_k \in D_{\delta_k}$, $d_k \to d^\star$, and $\limsup_k v(\delta_k) \le v(\delta)$. Conversely, choose minimizers in D_{δ_k} along a subsequence realizing the lower limit. Compactness gives a further subsequence converging to some $\bar{d} \in D_\delta$, and continuity of h gives $\liminf_k v(\delta_k) = h(\bar{d}) \ge v(\delta)$. At $\delta = 0$, every $d_k \in D_{\delta_k}$ converges to zero. Hence v is continuous on $[0, +\infty)$; the same argument applies to the envelope outer value. It follows that

$$0 < r_\lambda^{\mathrm{esc}}(x) < +\infty \quad \Longrightarrow \quad \mathcal{R}_{\lambda, r_\lambda^{\mathrm{esc}}(x)}(x) = 0.$$

When the radius is zero, the boundary value $\mathcal{R}_{\lambda,0}(x) = f(x) - f(P_\lambda^{\mathcal{C}}(x))$ can instead be zero or positive; it is zero whenever x is a proximal fixed point, as is the case for the KKT candidates considered below.

The classification begins with a basic consequence of proximal optimality. For every feasible center $z \in \mathcal{C}$, setting $p = P_\lambda^{\mathcal{C}}(z)$ and using $u = z$ as a competitor gives

$$f(p) + \frac{1}{2\lambda}\|p - z\|^2 \le f(z). \tag{5.4}$$

Inequality (5.4) shows that every direct feasible decrease yields at least as large a polished decrease. Define accordingly the direct feasible decrease

$$\mathcal{S}_\delta(x) := f(x) - \inf_{\substack{u \in \mathcal{C} \\ \|u - x\| \le \delta}} f(u). \tag{5.5}$$

For any feasible $u = x + d$ in the radius-δ ball, equation (5.4) gives $f(P_\lambda^{\mathcal{C}}(x+d)) \le f(u)$. Taking infima yields

$$\mathcal{R}_{\lambda,\delta}(x) \ge \mathcal{S}_\delta(x). \tag{5.6}$$

The positive-radius statement also requires stability of the safe proximal map. By Lemma 2.5, its Lipschitz constant is at most $(1 - \lambda\rho)^{-1}$, the relevant weakly convex stability bound.

Theorem 5.2 (Trichotomy for the ideal escape radius)**.** *Let $f + \delta_{\mathcal{C}}$ be proper, lower semicontinuous, and ρ-weakly convex, and let $\lambda > 0$ with $\lambda\rho < 1$. For a feasible point $x \in \operatorname{dom} f$, exactly one of the following cases holds.*

(i) x is a global minimizer of f on $\mathcal{C}$, and $r_\lambda^{\mathrm{esc}}(x) = +\infty$.

(ii) *x is a nonglobal local minimizer of f on $\mathcal{C}$, and $0 < r_\lambda^{\mathrm{esc}}(x) < +\infty$. More precisely, if $r_0 > 0$ satisfies*

$$f(u) \geq f(x) \qquad \text{for all } u \in \mathcal{C} \cap B_{r_0}(x),$$

then

$$r_\lambda^{\mathrm{esc}}(x) \geq (1 - \lambda\rho) r_0. \tag{5.7}$$

(iii) *x is not a local minimizer of f on $\mathcal{C}$, and $r_\lambda^{\mathrm{esc}}(x) = 0$.*

Equivalently, the three possible radius types $+\infty$, $(0, +\infty)$, and 0 characterize, respectively, the three classes above.

Proof. If x is globally minimizing, every proximal output is feasible and has objective value at least $f(x)$. The unperturbed center has proximal point x, so $\mathcal{R}_{\lambda,\delta}(x) = 0$ for every δ. Conversely, if a feasible z satisfies $f(z) < f(x)$, then equation (5.4) gives $f(P_\lambda^{\mathcal{C}}(z)) \leq f(z) < f(x)$. Taking $\delta \geq \|z - x\|$ proves that the escape radius is finite. Hence the infinite-radius case is equivalent to global minimality.

If x is not locally minimizing, every positive-radius feasible ball contains a point with smaller objective value. Equation (5.6) therefore makes the polished decrease positive at every radius, and the radius is zero.

It remains to consider a local minimizer x. The nonsmooth Fermat rule applied to $f + \delta_{\mathcal{C}}$ gives $0 \in \partial(f + \delta_{\mathcal{C}})(x)$ [22, Theorem 10.1]. By Lemma 2.5, this is equivalent in the safe regime to $P_\lambda^{\mathcal{C}}(x) = x$. Choose $r_0 > 0$ such that $f(u) \geq f(x)$ for all $u \in \mathcal{C} \cap B_{r_0}(x)$. Then equation (2.16) gives, whenever $\|d\| \leq (1 - \lambda\rho) r_0$,

$$\left\| P_\lambda^{\mathcal{C}}(x + d) - x \right\| \leq \frac{\|d\|}{1 - \lambda\rho} \leq r_0.$$

The proximal output is feasible and lies in the local-minimum neighborhood, so its value is at least $f(x)$. The choice $d = 0$ attains $f(x)$, hence the polished decrease is zero throughout this range and equation (5.7) follows. A local minimizer therefore has positive radius. If it is nonglobal, the already proved global-minimum equivalence makes its radius finite, hence strictly between zero and $+\infty$. These alternatives exhaust all feasible points. □

The trichotomy is value based. If the restriction of f to $\mathcal{C}$ is twice differentiable near a stationary point, a negative eigenvalue of the Hessian on $\mathcal{T}$ implies case (iii) by a Taylor expansion. A positive-semidefinite, or even zero, reduced Hessian may also coexist with higher-order feasible descent; such points have $r_\lambda^{\mathrm{esc}}(x) = 0$ because Theorem 5.2 uses realized nearby objective values.

A positive finite radius is a λ-dependent threshold in *center space*, followed by proximal polishing. Among the stationary candidates produced by the SDG, it distinguishes global minima, nonglobal local minima, and points that are not locally minimizing. Figure 5.1 illustrates both the two-stage mechanism and the three possible decrease profiles. Panel (a) uses the illustrative one-dimensional objective $f(t) = (t^2 - 1)^2/4 - 0.15t$, with $\rho = 1$ and $\lambda = 0.7$.

The direct decrease also induces a local feasible support curvature:

$$\begin{aligned} \beta_r^{\mathcal{C}}(x) &:= \sup_{\substack{u \in \mathcal{C} \cap \operatorname{dom} f \\ 0 < \|u - x\| \leq r}} \frac{2[f(x) - f(u)]_+}{\|u - x\|^2} \\ &= \inf \left\{ \beta \geq 0 : x \in \operatorname*{arg\,min}_{\substack{u \in \mathcal{C} \\ \|u - x\| \leq r}} \left(f(u) + \frac{\beta}{2} \|u - x\|^2 \right) \right\}, \end{aligned} \tag{5.8}$$

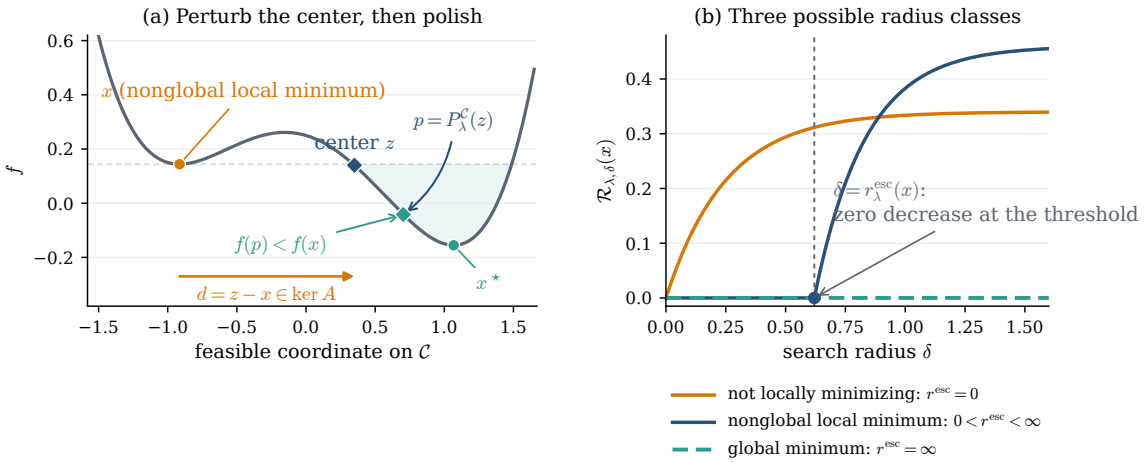


Fig. 5.1 Proximal-escape mechanism and radius classification. (a) A feasible center perturbation is followed by the safe constrained proximal map, producing the lower-value witness p. (b) The three profiles in Theorem 5.2; the positive-radius profile meets zero continuously at its threshold. Panel (b) is schematic.

where both expressions take values in $[0, +\infty]$. The supremum of an empty set is zero; if the quotient is unbounded, the admissible set in the infimum representation is empty and $\inf \varnothing = +\infty$. In particular, $\beta_r^{\mathcal{C}}(x) = 0$ if and only if $\mathcal{S}_r(x) = 0$. The multiplier-free quantity $\beta_r^{\mathcal{C}}(x)$ describes feasible descent in a fixed ball. If $\beta_r^{\mathcal{C}}(x) > 0$, then $\mathcal{S}_r(x) > 0$, and equation (5.6) implies $r_\lambda^{\mathrm{esc}}(x) \le r$. The local feasible quantity $\beta_r^{\mathcal{C}}(x)$ is distinct from the all-space Lagrangian threshold $\beta_{\mathrm{crit}}(x, y)$ in equation (4.4).

5.2 An envelope formulation

5.2.1 Three ordered decrease measures

The map $d \mapsto f(P_\lambda^{\mathcal{C}}(x+d))$ defining equation (5.1) need not be smooth, even though the safe proximal map is single valued. A smooth outer objective is obtained from the constrained Moreau envelope:

$$\mathcal{R}_{\lambda,\delta}^{E}(x) := f(x) - \inf_{\substack{d \in \mathcal{T} \\ \|d\| \le \delta}} \mathcal{E}_\lambda^{\mathcal{C}}(x+d). \tag{5.9}$$

Together with the direct and polished decreases, this gives three measures of improvement whose order determines the strength of their associated rejection tests.

Lemma 5.3 (Ordering of the three decrease measures)**.** *For every feasible x and every $\delta \ge 0$,*

$$\mathcal{S}_\delta(x) \le \mathcal{R}_{\lambda,\delta}^{E}(x) \le \mathcal{R}_{\lambda,\delta}(x). \tag{5.10}$$

Proof. For a feasible center $z = x + d$, using $u = z$ as a competitor gives $\mathcal{E}_\lambda^{\mathcal{C}}(z) \le f(z)$. If $p = P_\lambda^{\mathcal{C}}(z)$, then

$$f(p) = \mathcal{E}_\lambda^{\mathcal{C}}(z) - \frac{1}{2\lambda}\|p - z\|^2 \le \mathcal{E}_\lambda^{\mathcal{C}}(z).$$

Taking infima over the same tangent ball and subtracting from $f(x)$ reverses both inequalities. □

The second inequality in equation (5.10) makes every $\mathcal{R}_{\lambda,\delta}^{E}(x) > 0$ a valid rejection certificate. The same proximal point also gives

$$\nabla \mathcal{E}_\lambda^{\mathcal{C}}(z) = \frac{1}{\lambda}\left(z - P_\lambda^{\mathcal{C}}(z)\right), \qquad f(P_\lambda^{\mathcal{C}}(z)) = \mathcal{E}_\lambda^{\mathcal{C}}(z) - \frac{\lambda}{2}\left\|\nabla \mathcal{E}_\lambda^{\mathcal{C}}(z)\right\|^2. \tag{5.11}$$

The polished value is available at no additional inner-prox cost.

Each decrease induces a corresponding infimal positivity radius:

$$r_\lambda^P(x) := \inf\{\delta > 0 : \mathcal{R}_{\lambda,\delta}(x) > 0\}, \quad r_\lambda^E(x) := \inf\{\delta > 0 : \mathcal{R}^E_{\lambda,\delta}(x) > 0\},$$

$$r^D(x) := \inf\{\delta > 0 : \mathcal{S}_\delta(x) > 0\}.$$

All three definitions use the convention $\inf \varnothing = +\infty$. The nesting in equation (5.10) implies

$$r_\lambda^P(x) \le r_\lambda^E(x) \le r^D(x), \qquad r_\lambda^P(x) = r_\lambda^{\mathrm{esc}}(x). \tag{5.12}$$

The smooth envelope can detect an escape no earlier than exact proximal polishing and no later than a search restricted to the perturbed center. For a proximal fixed point x, this comparison has a complementary lower bound:

$$(1 - \lambda\rho) r^D(x) \le r_\lambda^P(x) \le r_\lambda^E(x) \le r^D(x). \tag{5.13}$$

Indeed, let $\mathcal{L}_x = \{u \in \mathcal{C} : f(u) < f(x)\}$, so that $r^D(x) = \operatorname{dist}(x, \mathcal{L}_x)$. Any polished escape $p = P_\lambda^{\mathcal{C}}(x + d) \in \mathcal{L}_x$ satisfies

$$r^D(x) \le \|p - x\| \le \frac{\|d\|}{1 - \lambda\rho}.$$

Taking the infimum over such perturbations proves the lower bound. If $\mathcal{L}_x$ is empty, all four quantities in equation (5.13) are infinite. Polishing therefore changes the center-space scale needed to discover decrease, not the underlying optimality classification.

5.2.2 A single-level representation

Interchanging the outer-center and inner-proximal minimizations yields the following single-level representation.

Lemma 5.4 (Soft-ball representation)**.** *For every feasible x and every $\delta \ge 0$,*

$$\inf_{\substack{d \in \mathcal{T} \\ \|d\| \le \delta}} \mathcal{E}_\lambda^{\mathcal{C}}(x + d) = \inf_{u \in \mathcal{C}} \left\{ f(u) + \frac{(\|u - x\| - \delta)_+^2}{2\lambda} \right\}. \tag{5.14}$$

Proof. Expanding the envelope and interchanging two infima gives

$$\inf_{\substack{u \in \mathcal{C},\ d \in \mathcal{T} \\ \|d\| \le \delta}} \left\{ f(u) + \frac{1}{2\lambda} \|u - x - d\|^2 \right\}.$$

Since $u - x \in \mathcal{T}$, the distance from $u - x$ to the tangent ball $B_\delta(0) \cap \mathcal{T}$ is $(\|u - x\| - \delta)_+$. Minimizing first in d proves equation (5.14). □

Equation (5.14) shows that the envelope search replaces the hard local ball in equation (5.5) by a quadratic penalty on distance outside that ball. The representation is exact, although the resulting global outer problem remains generally nonconvex.

Evaluating this envelope requires a constrained proximal point. We first describe two ways to obtain it, then combine these evaluations with a projected search and a finite rejection test.

5.3 Affine proximal oracles

Two exact reductions implement the hard-constrained proximal map while preserving affine feasibility. The first works directly in the tangent space; the second enforces the constraint by solving a dual root.

5.3.1 Tangent-space reduction

Let

$$P_{\mathcal{T}} := I - A^\top (AA^\top)^\dagger A \tag{5.15}$$

be the orthogonal projector onto $\ker A$, where † denotes the Moore–Penrose inverse. The orthogonal normal projector is $P_{\mathcal{T}^\perp} = I - P_{\mathcal{T}}$.

Lemma 5.5 (Normal invariance)**.** *For every $z, s \in \mathbb{R}^n$,*

$$P_\lambda^{\mathcal{C}}(z+s) = P_\lambda^{\mathcal{C}}(z + P_{\mathcal{T}} s). \tag{5.16}$$

Proof. Fix any $c \in \mathcal{C}$. Every $u \in \mathcal{C}$ can be written as $u = c + t$ with $t \in \mathcal{T}$. Orthogonal decomposition gives

$$\|u - (z+s)\|^2 = \|t - P_{\mathcal{T}}(z+s-c)\|^2 + \|P_{\mathcal{T}^\perp}(z+s-c)\|^2 .$$

The second term is independent of u, while

$$P_{\mathcal{T}}(z+s-c) = P_{\mathcal{T}}(z + P_{\mathcal{T}} s - c).$$

Hence the two proximal objectives differ on $\mathcal{C}$ only by an additive constant. Their unique minimizers are equal. □

Normal invariance also shows that restricting the outer search to the tangent space loses no polished values:

$$\inf_{\substack{s\in\mathbb{R}^n \\ \|s\|\le\delta}} f\Big(P_\lambda^{\mathcal{C}}(x+s)\Big) = \inf_{\substack{d\in\mathcal{T} \\ \|d\|\le\delta}} f\Big(P_\lambda^{\mathcal{C}}(x+d)\Big). \tag{5.17}$$

Indeed, $P_{\mathcal{T}} s$ has norm no larger than s and gives the same proximal point, while every tangent d is already an admissible full-space perturbation. Equation (5.17) shows that normal components are computationally redundant as escape directions.

If $N \in \mathbb{R}^{n\times k}$ has orthonormal columns spanning $\mathcal{T}$, every feasible point has the form $x + Nq$ for a fixed feasible x. For a tangent center $x + N\xi$,

$$P_\lambda^{\mathcal{C}}(x + N\xi) = x + N \arg\min_q \left\{ f(x+Nq) + \frac{1}{2\lambda}\|q-\xi\|^2 \right\}. \tag{5.18}$$

Indeed, $\|N(q-\xi)\| = \|q-\xi\|$, and injectivity of N makes the reduced minimizer unique. The formula removes the full-space affine condition algebraically. The nullspace formulation is attractive when $k = n - \operatorname{rank} A$ is small. It enforces feasibility exactly in the mathematical formulation and, in floating-point arithmetic, up to the tolerance of the nullspace linear algebra, without introducing a penalty parameter.

5.3.2 Dual-root reduction

When the unconstrained prox of f is easier than the nullspace problem, the affine constraint can instead be enforced through the dual-root construction of Hou et al. [21, Section 2.1], specialized here to the Euclidean root map

$$H_z(w) := A \operatorname{prox}_{\lambda f}(z + A^\top w) - b. \tag{5.19}$$

Proposition 5.6 (Exact affine-prox recovery from a dual root)**.** *Suppose f is proper, lower semicontinuous, and ρ-weakly convex, and $\lambda > 0$ with $\lambda\rho < 1$. If $H_z(w) = 0$, then*

$$p = \operatorname{prox}_{\lambda f}(z + A^\top w) \tag{5.20}$$

is the unique point $P_\lambda^{\mathcal{C}}(z)$.

Proof. The root condition makes p feasible. For every $u \in \mathcal{C}$,

$$\begin{aligned}\left\|u - (z + A^\top w)\right\|^2 - \|u - z\|^2 &= -2\langle A(u - z), w\rangle + \left\|A^\top w\right\|^2 \\ &= -2\langle b - Az, w\rangle + \left\|A^\top w\right\|^2,\end{aligned}$$

which is independent of u. The point p globally minimizes the shifted unconstrained proximal objective and belongs to $\mathcal{C}$; therefore it also minimizes that objective over $\mathcal{C}$. Removing the displayed constant shows that it minimizes the constrained proximal objective centered at z. Uniqueness follows from $\lambda\rho < 1$. □

Proposition 5.7 (Dual-root existence under affine qualification)**.** *In addition to the assumptions of Proposition 5.6, suppose that*

$$\mathcal{C} \cap \operatorname{ri}(\operatorname{dom} f) \neq \varnothing. \tag{5.21}$$

Then, for every center $z \in \mathbb{R}^n$, the equation $H_z(w) = 0$ has at least one solution.

Proof. Let $p = P_\lambda^{\mathcal{C}}(z)$. The function

$$u \longmapsto \lambda f(u) + \frac{1}{2}\|u - z\|^2$$

is proper, lower semicontinuous, and $(1-\lambda\rho)$-strongly convex. The relative-interior condition (5.21) is the standard qualification for its restriction to $Au = b$. The exact affine sum rule and convex Fermat rule therefore produce a multiplier η such that

$$0 \in \lambda\partial f(p) + p - z + A^\top\eta, \qquad Ap = b \tag{5.22}$$

by equation (2.6) and the Fermat rule [22, Theorem 10.1]. Set $w = -\eta$. The first inclusion in equation (5.22) is equivalent to

$$p = \operatorname{prox}_{\lambda f}(z + A^\top w),$$

because the shifted unconstrained proximal objective is strongly convex. The feasibility equation then gives $H_z(w) = 0$. □

The root map has an underlying convex potential. Define

$$h(u) := \lambda f(u) + \frac{1}{2}\|u\|^2, \qquad \Psi_z(w) := h^*(z + A^\top w) - \langle b, w\rangle .$$

The function h is $(1-\lambda\rho)$-strongly convex; hence h^* is differentiable and $\nabla h^*(q) = \operatorname{prox}_{\lambda f}(q)$. It follows that

$$\nabla\Psi_z(w) = H_z(w), \qquad \operatorname{Lip}(H_z) \le \frac{\|A\|^2}{1-\lambda\rho}. \tag{5.23}$$

Hence H_z is a monotone Lipschitz gradient field; convexity pertains to the scalar potential Ψ_z.

Any root in Proposition 5.6 recovers the affine proximal point; the qualification in Proposition 5.7 guarantees existence, while the multiplier w may be nonunique. The safe weak-convexity condition makes the shifted proximal point the unique global solution used by the escape certificate.

5.4 Numerical search and rejection

5.4.1 Projected envelope search

Using these proximal oracles, fix the KKT candidate x and a trial radius $\delta > 0$, and minimize $F(d) := \mathcal{E}^{\mathcal{C}}_\lambda(x+d)$ over $D_\delta = \mathcal{T} \cap B_\delta(0)$. Here the optimization variable is the center perturbation d, not the candidate x. Finding any $d \in D_\delta$ with $F(d) < f(x)$ is sufficient: the recovered point $p = P^{\mathcal{C}}_\lambda(x+d)$ then satisfies $f(p) \le F(d) < f(x)$ by equation (5.11).

Projected gradient descent provides one way of generating trial perturbations. At iteration k, a constrained proximal evaluation gives both $F(d_k)$ and its gradient. Since the center $z_k = x + d_k$ and the proximal output are feasible, their difference belongs to $\mathcal{T}$. Projecting the gradient step back onto D_δ retains the radius constraint:

$$\begin{aligned} z_k &= x + d_k, & p_k &= P^{\mathcal{C}}_\lambda(z_k), & g_k &= \lambda^{-1}(z_k - p_k), \\ d_{k+1} &= \Pi_{\mathcal{T}\cap B_\delta(0)}(d_k - \alpha_k g_k). \end{aligned} \tag{5.24}$$

To choose a stepsize, note that $\mathcal{E}^{\mathcal{C}}_\lambda$ is the Moreau envelope of the proper, lower-semicontinuous, ρ-weakly convex function $f + \delta_{\mathcal{C}}$. Standard weakly convex Moreau calculus therefore gives

$$L_E := \max\left\{\frac{1}{\lambda}, \frac{\rho}{1-\lambda\rho}\right\} \tag{5.25}$$

as a Lipschitz constant for its gradient [15, Lemma 2.5]. Specialize $\alpha_k \equiv \alpha$ with $0 < \alpha \le L_E^{-1}$. The projection variational inequality and the descent lemma give

$$\begin{aligned} F(d_{k+1}) &\le F(d_k) - \frac{1}{2\alpha}\|d_{k+1} - d_k\|^2 \\ &= F(d_k) - \frac{\alpha}{2}\|\mathcal{G}_\alpha(d_k)\|^2, \qquad \mathcal{G}_\alpha(d) := \alpha^{-1}\left[d - \Pi_{D_\delta}(d - \alpha\nabla F(d))\right]. \end{aligned} \tag{5.26}$$

For example, the first line follows from $\langle \nabla F(d_k), d_{k+1} - d_k\rangle \le -\alpha^{-1}\|d_{k+1} - d_k\|^2$, followed by the L_E-smooth descent bound. The argument is the classical gradient-projection mechanism [18], here applied to the safe envelope. This descent estimate uses an exact proximal point. The following bound records the effect of an inexact inner solve.

Lemma 5.8 (Inexact projected-envelope step)**.** *Set* $z = x + d$, $\mu_\lambda = \lambda^{-1} - \rho > 0$, *and* $Q_z(u) = f(u) + \delta_{\mathcal{C}}(u) + \|u - z\|^2/(2\lambda)$. *Suppose* $\operatorname{dist}(0, \partial Q_z(\widehat{p})) \le \varepsilon$, *and let* $\widehat{g} = (z - \widehat{p})/\lambda$. *Then*

$$\left\|\widehat{p} - P_\lambda^{\mathcal{C}}(z)\right\| \le \frac{\varepsilon}{\mu_\lambda}, \qquad 0 \le Q_z(\widehat{p}) - \mathcal{E}_\lambda^{\mathcal{C}}(z) \le \frac{\varepsilon^2}{2\mu_\lambda},$$
$$\|\widehat{g} - \nabla F(d)\| \le \frac{\varepsilon}{1 - \lambda\rho}. \tag{5.27}$$

For $d^+ = \Pi_{D_\delta}(d - \alpha\widehat{g})$, *with* $0 < \alpha \le L_E^{-1}$,

$$F(d^+) \le F(d) - \frac{1}{4\alpha}\left\|d^+ - d\right\|^2 + \frac{\alpha\varepsilon^2}{(1 - \lambda\rho)^2}. \tag{5.28}$$

Proof. Apply the strong-convexity argument of Lemma 3.4 to Q_z; the gradient estimate follows by dividing the point error by λ. Writing $s = d^+ - d$, the projection inequality and L_E-smoothness give

$$F(d^+) \le F(d) - \frac{\|s\|^2}{2\alpha} + \|\widehat{g} - \nabla F(d)\|\,\|s\|.$$

Use $ab \le a^2/(4\alpha) + \alpha b^2$ to conclude. □

The residual here includes the constraint normal cone. In particular, the hypothesis requires a feasible inner point; a small dual-root feasibility residual alone is not this bound. A fixed inner tolerance permits an additive error in descent, rather than the exact monotonicity of equation (5.26).

At a KKT candidate, $P_\lambda^{\mathcal{C}}(x) = x$, so initialization at $d_0 = 0$ would leave the iteration at zero. Nonzero initial perturbations, possibly with restarts, are therefore needed. Descent of F does not by itself guarantee that its value falls below $f(x)$; the envelope and polished values must be tested against that fixed reference value.

A finite collection of the generated perturbations can be used as the set $\mathcal{D}$ in Algorithm 5.1. That algorithm handles the rejection decision, whereas equation (5.24) supplies trial centers. The proximal evaluations use the reductions in Section 5.3; the rejection test below also addresses inexact inner solves.

5.4.2 Warm starts and dual-root stopping

Let (x, ϕ) be a KKT pair of equation (1.1). At the unperturbed center, the exact dual root is

$$w_x = -\lambda\phi, \tag{5.29}$$

because

$$x = \operatorname{prox}_{\lambda f}(x - \lambda A^\top \phi).$$

For a perturbed center $x + s$, define the normal correction

$$q_s := (AA^\top)^\dagger As, \qquad w^{(0)}(s) := -\lambda\phi - q_s. \tag{5.30}$$

Then

$$x + s + A^\top w^{(0)}(s) = x + P_{\mathcal{T}} s - \lambda A^\top \phi. \tag{5.31}$$

A purely normal change in the center is corrected exactly. For a tangent change, equation (5.31) gives the identity

$$H_{x+s}(w^{(0)}(s)) = H_{x+P_{\mathcal{T}}s}(-\lambda\phi). \tag{5.32}$$

Since $H_x(-\lambda\phi) = 0$, Lipschitz continuity of the safe unconstrained prox then gives the initial residual bound

$$\left\|H_{x+s}(w^{(0)}(s))\right\| \le \frac{\|A\|}{1-\lambda\rho}\,\|P_{\mathcal{T}}s\|\,. \tag{5.33}$$

Remark 5.9 (Scope of the residual bound). Equation (5.33) controls the initial residual. Distance to a selected multiplier additionally requires metric regularity or strong monotonicity because multipliers may be nonunique and nonsmooth proximal maps may generate an interval of dual roots.

The same potential supplies a safeguard for a basic root iteration. Let

$$L_H := \frac{\|A\|^2}{1-\lambda\rho}$$

and, when $L_H > 0$, take $0 < \eta \le L_H^{-1}$. The gradient step

$$w^+ = w - \eta H_z(w) \tag{5.34}$$

satisfies

$$\Psi_z(w^+) \le \Psi_z(w) - \frac{\eta}{2}\,\|H_z(w)\|^2\,. \tag{5.35}$$

Indeed, equation (5.23) and the smooth descent lemma yield the stronger coefficient $\eta(1 - \eta L_H/2) \ge \eta/2$. Backtracking may enforce the displayed decrease without knowing a sharp Lipschitz constant.

If $\widetilde{p} = \mathrm{prox}_{\lambda f}(z + A^\top w)$, then

$$\|H_z(w)\| = \|A\widetilde{p} - b\| \tag{5.36}$$

is exactly the affine feasibility residual of the shifted unconstrained proximal point. Together with descent of Ψ_z, it is a natural stopping diagnostic. Translating it into objective-value accuracy additionally requires control of the dual-root error and of any inexact unconstrained prox computation.

5.4.3 Finite rejection certificates

A finite search retains the one-sided validity of the ideal definition. Given centers $d_1, \dots, d_N \in \mathcal{T} \cap B_\delta(0)$, let

$$\widehat{\mathcal{R}}^{(N)}_{\lambda,\delta}(x) := \max_{1\le i\le N}\left[f(x) - f\left(P^{\mathcal{C}}_\lambda(x+d_i)\right)\right]_+\,. \tag{5.37}$$

By construction,

$$0 \le \widehat{\mathcal{R}}^{(N)}_{\lambda,\delta}(x) \le \mathcal{R}_{\lambda,\delta}(x). \tag{5.38}$$

Hence a positive sampled value certifies rejection when the inner affine proximal points are exact or accompanied by rigorous feasible value bounds. With floating-point values, the comparison should require

$$f(u) \le f(x) - \tau_{\mathrm{cert}},$$

where $\tau_{\mathrm{cert}} > 0$ dominates a certified bound on the combined evaluation and inner-solve error. A zero sampled value does not certify the absence of an unsampled escape. Equivalently, if validated objective enclosures are available, the test is

$$f(x) \in [\underline{f}_x, \overline{f}_x], \quad f(u) \in [\underline{f}_u, \overline{f}_u], \quad \overline{f}_u < \underline{f}_x,$$

for a certified feasible u. A small floating-point feasibility residual alone does not establish the existence of such a feasible witness.

An inexact nullspace computation may still return an explicitly feasible point $u = x + Nq$. A certified inequality $f(u) < f(x)$ then gives a direct rejection witness independently of reduced-proximal optimality; identifying it with the sampled proximal value in equation (5.37) additionally requires inner optimality. A dual iterate with $H_z(w) \neq 0$ is infeasible, so its objective value becomes admissible only after obtaining a certified feasible point or a problem-specific error enclosure. Algorithm 5.1 summarizes the decision logic.

Algorithm 5.1 Finite proximal-escape search with certified rejection

Require: KKT candidate $x \in \mathcal{C}$; parameter $\lambda > 0$ with $\lambda\rho < 1$; radius $\delta > 0$; finite set $\mathcal{D} \subset \mathcal{T} \cap B_\delta(0)$
Ensure: REJECTED with a feasible witness, or UNRESOLVED
1: **for** $d \in \mathcal{D}$ **do**
2: $z \leftarrow x + d$
3: attempt to compute $P_\lambda^{\mathcal{C}}(z)$ by equation (5.18) or by a certified root of equation (5.19)
4: **if** the inner computation certifies a point $u \in \mathcal{C}$ and $f(u) < f(x)$ **then**
5: **return** REJECTED with witness u
6: **end if**
7: **end for**
8: **return** UNRESOLVED

Directional coverage remains a limitation in high dimension. If an escape cone has relative spherical measure p, then N independent uniform directions hit it with probability $1 - (1-p)^N$. Narrow cones can have rapidly vanishing p as the tangent dimension grows. Adaptive directions, structured subspace searches, or problem-specific deterministic directions are therefore needed as the tangent dimension grows.

Replacing $Au = b$ in equation (2.14) by a penalty $\gamma \|Au - b\|^2 / 2$ produces a different problem for every finite γ. An infeasible point may have lower raw objective than every feasible point and can therefore create a false rejection even when its constraint residual is small. Using such an output as an exact feasible rejection certificate requires an error analysis that converts infeasibility and objective error into a valid feasible bound.

6 Numerical experiments

We first compare the ability of the safe SDG and two residual merits to recover KKT points, then examine what proximal escape reveals about the recovered candidates. Two small problems with complete stationary catalogs make missed branches identifiable. The second problem also has analytic escape radii, allowing the distinction between stationarity and minimality to be checked without relying on an outer numerical optimizer.

6.1 Recovering stationary branches

6.1.1 Test problems

The affine rank-one problem has a smooth objective and isolated KKT pairs:

$$\min_{v_1^\top x = c} f(x) = \frac{1}{4} \left\| xx^\top - M \right\|_F^2, \qquad M = V \operatorname{Diag}(d_1, \ldots, d_8) V^\top, \qquad c = 0.4, \tag{6.1}$$

where $V = [v_1, \dots, v_8]$ is orthogonal and $d_i = 4 - (i-1)/2$. In the eigenbasis, stationarity in a free coordinate requires $(\|x\|^2 - d_j)x_j = 0$. Since the eigenvalues are distinct, at most one free coordinate is nonzero. The complete catalog is therefore

$$\begin{aligned} x_0 &= cv_1, & y_0 &= (d_1 - c^2)c, \\ x_j^{\pm} &= cv_1 \pm \sqrt{d_j - c^2}\, v_j, & y_j^{\pm} &= (d_1 - d_j)c, \quad j = 2, \dots, 8. \end{aligned} \tag{6.2}$$

At x_0, the Hessian restricted to $\ker A$ has eigenvalues $c^2 - d_k < 0$, $k \geq 2$. At $x_j^{\pm}$, they are $2(d_j - c^2)$ and $d_j - d_k$, $k \geq 2$, $k \neq j$. Thus x_0 is a strict constrained local maximum, the twelve points with $j \geq 3$ are strict saddles, and the two points with $j = 2$ are minima. They are global: at a fixed norm of the free component, the Rayleigh bound selects v_2, and minimization over that norm gives $\sqrt{d_2 - c^2}$. For the reported run, V is obtained by QR factorization of an 8×8 standard Gaussian matrix from NumPy's `SeedSequence([20260709,1401])`, with the signs chosen so that the diagonal of R is positive.

The box problem introduces active constraints and nonglobal local minima:

$$\min_{\substack{x \in [0,1]^7 \\ x_1 = 0.37}} F(x) = \sum_{i=1}^{7} F_i(x_i), \qquad F_i(t) = \frac{1}{2} q_i t^2 + r_i t. \tag{6.3}$$

Here $B = [0,1]^7$, $f = F + \delta_B$, and $\mathcal{C} = \{x : x_1 = 0.37\}$; hence $A = e_1^\top$ and $b = 0.37$, where e_i denotes the ith standard coordinate vector. The quadratic coefficients satisfy $q_2, q_3, q_4 < 0$, while all other q_i are positive. Coordinates $2, 3, 4$ are therefore concave, with interior maxima $a_i = -r_i/q_i$ given by $(a_2, a_3, a_4) = (0.30, 0.335, 0.37)$. The remaining free coordinates are strictly convex with interior minima. Each concave coordinate has three KKT states, $0, a_i, 1$, denoted by L, I, U, as illustrated in Figure 6.1. This gives $3^3 = 27$ KKT points: 19 constrained strict saddles and eight local minima. Here a strict saddle has a negative-curvature direction tangent to its active face, supplied by any interior concave coordinate. Among the endpoint combinations, UUU is the unique global minimum because

$$F_i(1) - F_i(0) = q_i(1/2 - a_i) < 0, \qquad i = 2, 3, 4.$$

The other seven local minima cannot be rejected by an arbitrarily small perturbation. This feature makes the box problem useful for the radius comparison in Section 6.2.

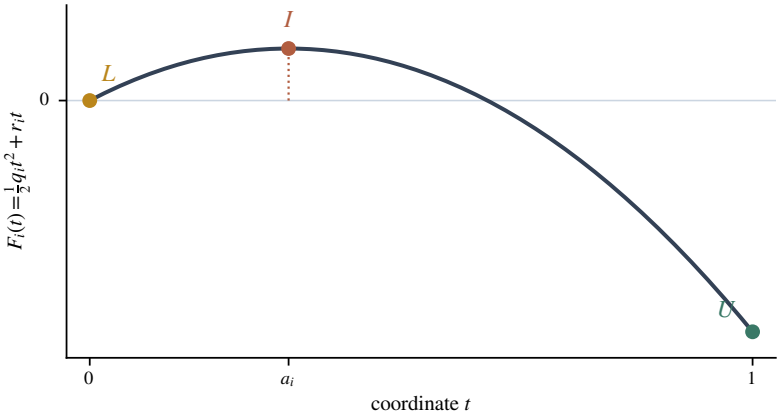


Fig. 6.1 Three KKT states of a concave box coordinate, illustrated for $i = 2$. The endpoints L and U are one-sided local minima; I is an interior maximum. Since $a_i < 1/2$, U is globally minimizing. Choosing one state for each of coordinates $2, 3, 4$ gives the box catalog.

Both problems admit inexpensive safe proximal evaluations. Write

$$p(x,y) = p_{\beta_x}(x,y) = \arg\min_u \left\{ f(u) + \langle y, Au - b\rangle + \frac{\beta_x}{2}\|u - x\|^2 \right\}. \tag{6.4}$$

For the rank-one objective, $\nabla^2 f(x) = 2xx^\top + \|x\|^2 I - M \succeq -4I$; we use $\rho = 4$ and $\beta_x = 5$. With $z = x - \beta_x^{-1} y v_1$, $\widehat{z} = V^\top z$, and $\widehat{p} = V^\top p(x,y)$, its evaluation reduces to

$$\sum_{i=1}^{8} \left(\frac{\beta_x \widehat{z}_i}{\beta_x + s - d_i} \right)^2 - s = 0, \qquad \widehat{p}_i = \frac{\beta_x \widehat{z}_i}{\beta_x + s - d_i}. \tag{6.5}$$

Writing the scalar left-hand side as $\varphi(s)$, one has $\varphi'(s) < -1$ for a nonzero center. Starting from $[0, \max\{1, \|z\|^2\}]$, the upper endpoint is doubled until $\varphi \le 0$, then Brent's method is applied with absolute tolerance 10^{-13} and relative tolerance $2 \cdot 10^{-14}$; the zero center returns zero directly. For an approximate root s, its point $p(s) = V\widehat{p}(s)$ satisfies $\nabla q_{x,y}(p(s)) = \varphi(s)p(s)$, so the scalar residual yields the inner stationarity residual used in Lemma 3.4. For the box objective $f = F + \delta_B$, we use $\rho = -\min_i q_i \simeq 2.24$ and $\beta_x = 3$, giving

$$p_{B,i}(x,y) = \Pi_{[0,1]} \left(\frac{\beta_x x_i - r_i - y\,\mathbf{1}_{\{i=1\}}}{\beta_x + q_i} \right). \tag{6.6}$$

This map enforces the box but dualizes the equality. The escape test, in contrast, will enforce both constraints in its proximal evaluation.

6.1.2 Merit comparison and branch coverage

We compare the SDG with the squared Lagrangian-proximal and KKT residuals:

$$\begin{aligned} M_{\mathrm{SDG}}(x,y) &= \mathcal{G}_{\beta_x,\beta_y}(x,y), \\ M_{\mathrm{prox}}(x,y) &= \frac{1}{2}\left(\beta_x^2 \|x - p(x,y)\|^2 + \|Ax - b\|^2 \right), \\ M_{\mathrm{KKT}}(x,y) &= \frac{1}{2}\left(\left\|\nabla f(x) + A^\top y\right\|^2 + \|Ax - b\|^2 \right). \end{aligned} \tag{6.7}$$

For the box problem, M_{prox} uses p_B, while the KKT stationarity residual is replaced by the natural mapping

$$N_B(x,y) = x - \Pi_B(x - \nabla F(x) - e_1 y).$$

The self-centered SDG (SC-SDG) uses the smooth finite extension in Proposition 3.6, minimized over $x \in B$, $y \in \mathbb{R}$. These are the actual outer bounds: the primal variables stay in the box, whereas the multiplier is unbounded. All three merits have the same KKT zero set; the comparison concerns how readily local minimization reaches different parts of that set.

Each merit is minimized with L-BFGS-B [23, 24], using $\beta_y = 1$ and identical starts within each problem. The affine comparison uses 160 isotropic primal starts with log-uniform radii in $[0.08, 3.2]$ and Gaussian multipliers of standard deviation 1.8. The box comparison uses 256 scrambled Sobol' points [25], generated with SciPy's linear-matrix scramble and digital shift, in $[0,1]^7 \times [-4,4]$; this multiplier interval applies only to initialization. Affine iterates are unbounded in both variables. The master seed is 20260709. The catalogs are used only to assess endpoints, not to initialize the searches. Analytic gradients are supplied for the smooth merits; the other box merits use piecewise-analytic derivatives, with clipping derivative zero at a transition. We use CPython 3.14.0, NumPy 2.3.5 and SciPy 1.16.3, with `ftol=1e-15` and `gtol=2e-10`. The iteration limits are 320 and 1000, and the line-search limits are 40 and 60, respectively; the affine run also sets `maxfun=320`.

Endpoints are accepted only if independently recomputed, normalized feasibility, stationarity, and proximal residuals are at most 10^{-7}; the box test also checks the hard-constrained proximal residual. In the affine case, these three quantities are

$$\frac{\|Ax-b\|}{1+\|b\|}, \qquad \frac{\left\|\nabla f(x)+A^\top y\right\|}{1+\|\nabla f(x)\|+\|A^\top y\|}, \qquad \frac{\beta_x\|x-p(x,y)\|}{1+\beta_x\|x\|+\|A^\top y\|}. \tag{6.8}$$

For the box, the four tests use

$$\frac{|x_1-0.37|}{1.37}, \quad \frac{\|N_B(x,y)\|}{1+\|x\|}, \quad \frac{\beta_x\|x-p_B(x,y)\|}{1+\beta_x\|x\|}, \quad \frac{\beta_x\left\|x-P^{\mathcal{C}}_{1/\beta_x}(x)\right\|}{1+\beta_x\|x\|}. \tag{6.9}$$

The box normalizations depend only on the primal point, so increasing the multiplier cannot suppress a saturated projection residual. Acceptance does not depend on the optimizer's success flag. An accepted endpoint is assigned to its nearest catalog pair in joint Euclidean (x,y) distance, with tolerance $8\cdot10^{-5}(1+\|(x,y)\|)$ at the affine endpoint and absolute tolerance $2\cdot10^{-5}$ for the box problem. Repeated matches count once in branch coverage. Every accepted endpoint in this run also matches its catalog; there are no accepted unmatched endpoints.

Table 6.1 KKT recovery from common starts. Accepted endpoints pass the independent residual tests; distinct roots count different catalog matches.

Family	Search objective	Accepted endpoints	Distinct roots
affine rank one	SC–SDG	159/160	15/15
	Lagrangian prox residual	160/160	15/15
	KKT residual	160/160	15/15
box QP	SC–SDG	101/256	27/27
	Lagrangian prox residual	109/256	10/27
	natural KKT residual	33/256	16/27

In this run, all three merits recover the fifteen affine KKT pairs (Table 6.1). On the box problem, SC-SDG recovers all 27 branches, compared with 10 for the proximal residual and 16 for the natural KKT residual. All methods find the eight local minima; their saddle coverage is respectively 19/19, 2/19, and 8/19. The proximal residual accepts more endpoints than SC-SDG but finds fewer branches, illustrating why endpoint success alone is not a sufficient measure of coverage.

The box result favors SC-SDG for this common-start L-BFGS-B comparison. It also reflects a regularity difference: unlike the other two box merits, SC-SDG does not require differentiating the clipping map. The comparison uses one seed and one Sobol scramble, so the observed coverage reflects both the merit and its interaction with L-BFGS-B. The endpoint counts are also sensitive to floating-point details: the run in Table 6.1 was made on Windows 11 with CPython 3.14, and with the same pinned NumPy, SciPy and Matplotlib versions on macOS (arm64, CPython 3.11) one affine SC-SDG start (start 45) ends just inside the acceptance tolerance, giving 160/160 instead of 159/160; all 768 box endpoints and every distinct-root count are unchanged.

6.2 Distinguishing the recovered candidates

6.2.1 Exact escape radii on the box problem

We now assess the exact analytic representatives of the matched box branches, not the floating-point endpoints themselves. The radii below are properties of these catalog points.

Descent from an approximate endpoint rejects only that endpoint; a small matching distance does not transfer the conclusion to its analytic branch. For example, for $f(t) = t^2$, the exact minimum 0 has infinite escape radius, while every nonzero nearby point has radius zero. For $\lambda = 1/3$, the hard-constrained proximal map is

$$P_\lambda^{\mathcal{C}}(z)_1 = 0.37, \qquad P_\lambda^{\mathcal{C}}(z)_i = \Pi_{[0,1]}\left(\frac{z_i - \lambda r_i}{1 + \lambda q_i}\right), \quad i = 2, \dots, 7. \tag{6.10}$$

Since $\lambda\rho \simeq 0.747 < 1$, the denominators are positive and the proximal point is unique. For a catalog point x, let

$$I(x) = \{i \in \{2,3,4\} : x_i = a_i\}, \qquad L(x) = \{i \in \{2,3,4\} : x_i = 0\}.$$

Proposition 6.1 (Exact escape radii for the box benchmark)**.** *The escape radius of every KKT point of equation (6.3) is*

$$r_\lambda^{\mathrm{esc}}(x) = \begin{cases} 0, & I(x) \neq \varnothing, \\ \min\limits_{i \in L(x)} a_i(2 + \lambda q_i), & I(x) = \varnothing,\ L(x) \neq \varnothing, \\ +\infty, & I(x) = L(x) = \varnothing. \end{cases} \tag{6.11}$$

Proof. An arbitrarily small perturbation of an interior concave coordinate is mapped away from its scalar maximum and decreases F, proving the first case. At a lower-bound state, the scalar quadratic first reaches an equally good point when the proximal output reaches $2a_i$. Solving $P_\lambda^{\mathcal{C}}(x + \tau_i e_i)_i = 2a_i$ gives $\tau_i = a_i(2 + \lambda q_i)$. Before this threshold, $0 \le p_i \le 2a_i$ and

$$F_i(p_i) - F_i(0) = q_i p_i \left(\frac{p_i}{2} - a_i\right) \ge 0$$

because $q_i < 0$. Upper-state concave coordinates and the convex coordinates are already at their scalar global minima. Thus separability rules out a cooperative decrease below the smallest threshold, whereas a perturbation along a minimizing coordinate produces strict decrease above it. If all concave coordinates are upper-state, every scalar term is globally minimized, giving the third case. □

Thus the 19 strict saddles have zero escape radius, the seven nonglobal local minima have positive finite radii, and the global minimum has infinite radius. The three possible finite thresholds are approximately 0.400, 0.433, and 0.464.

Figure 6.2 displays one representative of each class. In IUU and LUU, only coordinate 2 can lower the objective, so the full-ball decrease is obtained from its scalar proximal formula. For IUU, every positive radius produces a decrease. For LUU, the decrease remains zero up to $a_2(2 + \lambda q_2)$, then becomes positive. The UUU profile stays zero at every radius.

The analytic escape test therefore rejects all 26 nonglobal catalog points, verifying the radius classification in Theorem 5.2 through separable scalar calculations.

6.2.2 A stationary point with vanishing Hessian

The box saddles already have negative-curvature directions. To examine a case where a Hessian test is inconclusive, consider on $\mathbb{R}^{12}$

$$q_3(x) = 1 + \left(\|x\|^2 - 1\right)^3, \qquad \bar{x}_3 = e_1, \qquad q_{\min}(x) = \|x\|^4, \qquad x^\star_{\min} = 0. \tag{6.12}$$

Both base points have zero gradient and zero Hessian. Nevertheless,

$$q_3(e_1) - q_3((1-\delta)e_1) = (2\delta - \delta^2)^3 > 0 \qquad (0 < \delta < 2), \tag{6.13}$$

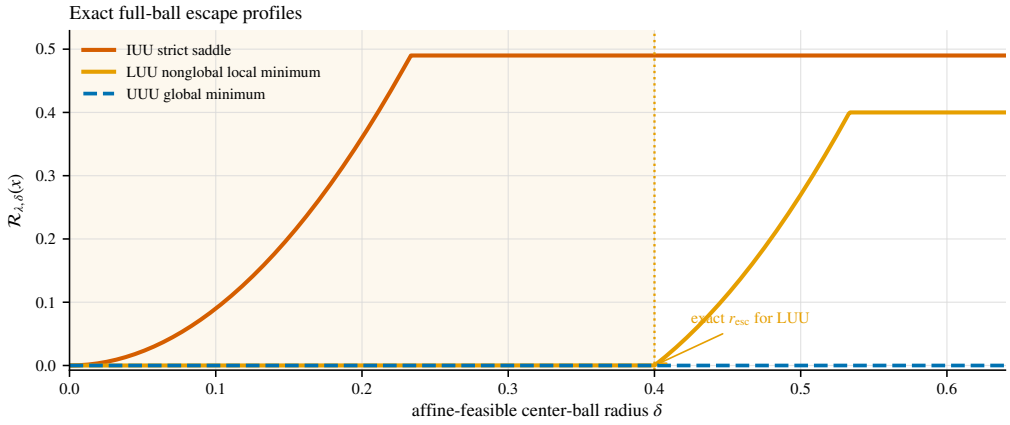


Fig. 6.2 Exact polished-decrease profiles for IUU, LUU, and UUU. Their escape radii are zero, approximately 0.400, and infinite, respectively; L, I, U denote the lower, interior, and upper states.

whereas $q_{\min}(x) \geq 0 = q_{\min}(0)$ for every x. Theorem 5.2 therefore gives escape radius zero for $\bar{x}_3$ and infinite radius for the flat minimum, although their first- and second-order data agree.

The radial and tangential Hessian eigenvalues of q_3, with $s = \|x\|^2$, are $6(s-1)(5s-1)$ and $6(s-1)^2$. Their lower bound is $-24/5$, so we take $\lambda = 1/8$, giving $\lambda\rho = 0.6$. We evaluate ten logarithmically spaced radii between 10^{-4} and 10^{-1}, with centers $z_j = (1-\delta_j)e_1$. The relevant proximal model is radial. More precisely, each center z_j and the objective are invariant under every orthogonal map fixing e_1; uniqueness in the safe regime forces the proximal point to lie in $\operatorname{span}\{e_1\}$. Its scalar coordinate t is consequently the unique bracketed root of

$$6t(t^2-1)^2 + 8(t-1+\delta) = 0. \tag{6.14}$$

The derivative of its left-hand side is at least $8 - 24/5 = 16/5$. Its signs at 0 and $1-\delta$ are opposite, so $0 < t_\delta < 1-\delta$ for the radii considered. We use this bracket in Brent's method, with absolute and relative tolerances 10^{-14} and $2 \cdot 10^{-14}$. To avoid subtracting objective values close to one, the plotted decreases are evaluated as

$$\begin{aligned} &\Delta_D = (2\delta - \delta^2)^3, \qquad \Delta_P = ((1-t_\delta)(1+t_\delta))^3, \\ &\Delta_E = \Delta_P - 4(t_\delta - 1 + \delta)^2. \end{aligned} \tag{6.15}$$

The control $q_{\min}$ rules out positive decrease analytically for every $\lambda > 0$, without an inner solve.

The direct, envelope, and polished decreases in Figure 6.3 follow the prescribed radial direction. Their positive sign follows from equation (6.13) and proximal optimality, independently of the plotted floating-point values. Values below $100\epsilon_{\mathrm{mach}}(1 + |q_3(e_1)|)$ are omitted as a reporting convention, not as an error certificate.

Along the analytically supplied direction, proximal escape detects nonminimality despite a vanishing Hessian.

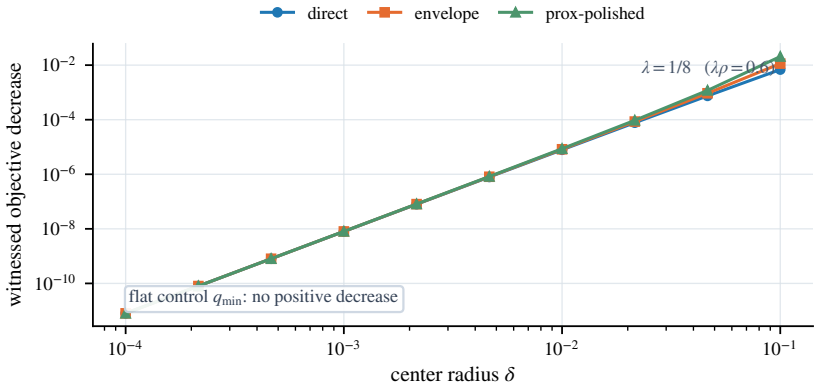


Fig. 6.3 Direct, envelope, and polished decreases along $z = (1 - \delta)e_1$, with $\lambda = 1/8$. The base point has zero gradient and Hessian. Curves are pathwise witnesses, not optimized full-ball values.

7 Discussion and conclusion

The SDG results separate a uniform computational regime from an exact pointwise characterization. If $\beta_x > \rho$, the primal comparison problem is strongly convex, the self-centered SDG vanishes exactly at KKT pairs, and its value controls feasibility and proximal displacement. At an arbitrary positive curvature, zero gap is equivalent to global quadratic support of the fixed Lagrangian, measured by $\beta_{\mathrm{crit}}(x, y)$. The threshold can vary across KKT branches and explains the false numerical zeros produced by a nonglobal local inner solution. The stronger zero interpretation of the global augmented-Lagrangian gap requires a global nonconvex value problem.

For post-KKT assessment, proximal escape uses a second safe construction: the hard-constrained proximal map of $f + \delta_{\mathcal{C}}$, distinct from the Lagrangian proximal comparison in the SDG. Its ideal radius is infinite at a global minimizer, finite and positive at a nonglobal local minimizer, and zero at a point that is not locally minimizing. Complete stationary catalogs assess branch recovery, the separable box problem verifies the radius trichotomy, and the Hessian-flat illustration shows that realized value decrease can reject stationary points with vanishing gradient and Hessian.

The conclusions assume global ρ-weak convexity, affine constraints, and reliable evaluation of the relevant safe proximal subproblems. Computing $\beta_{\mathrm{crit}}(x, y)$ remains a global problem. The finite rule in Algorithm 5.1 accepts a verified lower feasible value as a rejection certificate and records every other finite-search outcome as unresolved.

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